UNIVERSITY
OF OSLO

Master's thesis

Numerical methods for nonlocal conservation laws

Filmon Tesfamikael Misgane

Mathematics for Applications
60 ECTS study points

Department of Mathematics
Faculty of Mathematics and Natural Sciences

Spring 2026

Filmon Tesfamikael Misgane

Numerical methods for nonlocal conservation laws

Supervisor:
Ulrik Skre Fjordholm

Abstract

In this thesis we study finite volume schemes for a nonlocal traffic flow model and their relation to the corresponding local traffic flow model. We design first-order Lax–Friedrichs-type and Godunov-type schemes, discuss their stability properties, and investigate their convergence numerically. The experiments indicate that normalized quadrature weights are important for consistency in the nonlocal-to-local limit, and that the Godunov-type scheme gives smaller L^1 errors than the Lax–Friedrichs-type scheme. We also formulate a second-order Godunov-type scheme based on piecewise linear reconstruction and SSP Runge–Kutta time stepping, and compare its numerical performance with the first-order scheme.

Contents

List of Figures

List of Tables

Preface

This thesis is intended for readers with a solid foundation in real analysis and numerical methods for partial differential equations. All numerical methods developed in this thesis have been implemented in Python using object-oriented programming. The code is available in the folder `src` at: https://github.com/Filmon492/MAT5960

Acknowledgments. Thanks to the University of Oslo. The main acknowledgment goes to my supervisor, Professor Ulrik Skre Fjordholm. Thank you to my fellow students, in particular Simon Lederhilger and August Femtehjell, for all the support they provided me throughout this thesis. Special thanks to my family, who believed in me. I would also like to acknowledge the support of my friends: Jonas Beyan, Samsom Amaniel, Beletse Sbhatu, and Filmon Mehari.

Part I

Introduction and Background

Chapter 1

Introduction

Some important conservation laws arising in applications, such as traffic flow, are nonlinear. For such problems, it is often impossible to obtain explicit solution formulas. Hence, we rely on numerical methods to approximate the solutions as accurately and efficiently as possible. When designing such methods, it is natural to ask whether the resulting approximations are reliable, stable, and convergent.

A central step in designing numerical methods for PDEs, in particular conservation laws, is to understand the properties of the corresponding analytical solutions. These properties guide the consistency, stability, and convergence analysis of the numerical schemes. This thesis focuses on two traffic-flow models: a local model and a nonlocal model. The local model depends only on the traffic density at the current position, whereas the nonlocal model takes into account traffic information over a road segment of length $\epsilon > 0$ ahead of the car's current location. The relationship between these two models is discussed in detail in Chapter 2.

In this thesis, we study numerical methods for the nonlocal traffic-flow model and their relation to the corresponding local traffic-flow model. Following [8], we develop first-order finite volume schemes in Chapter 3 and study their stability properties in Chapter 4. The numerical experiments in Chapter 5 indicate first-order convergence, which motivates the search for higher-order methods. Following [6], we formulate a second-order Godunov scheme in Chapter 6 and compare its numerical performance with the first-order Godunov scheme in Chapter 7.

Chapter 2

Preliminaries

In this chapter, we present the necessary background on local and nonlocal conservation laws.

2.1 Scalar Local and Nonlocal Conservation Laws

Scalar conservation laws are partial differential equations that can be written in the form

$$\begin{aligned} u_t + f(u)_x &= 0, \qquad x \in \mathbb{R}, t > 0 \\ u(x,0) &= u_0(x), \qquad x \in \mathbb{R}, t = 0, \end{aligned} \tag{2.1}$$

where $u = u(x,t)$ is the unknown function, while the flux function f and the initial data $u_0(x)$ are given.

If we now let $f(u) = uv(u)$ where $v(u) = 1 - u$, then Eq. (2.1) can be used for modelling traffic flow. For a detailed setup of this model, we refer to [5]. This model is however based only on local information. When a car driver usually decides his speed, it depends on the traffic information within a road segment of length $\epsilon > 0$ ahead of the car's current location. For managing that case we look for a better way to model the velocity, so we need a nonlocal model which is in the form

$$\begin{aligned} \rho_t + (\rho V_\epsilon(\rho))_x &= 0, \qquad x \in \mathbb{R}, t > 0 \\ \rho(x,0) &= \rho_0(x), \qquad x \in \mathbb{R}, t = 0, \end{aligned} \tag{2.2}$$

where now V_ϵ is given as

$$\begin{aligned} V_\epsilon(\rho)(x,t) &= v(\rho * \omega^\epsilon) \\ &= 1 - \int_0^\epsilon \omega^\epsilon(y)\rho(x+y,t)\,dy \\ &= 1 - \int_x^{x+\epsilon} \omega^\epsilon(y-x)\rho(y,t)\,dy \qquad \text{Change of variable} \end{aligned} \tag{2.3}$$

for some nonlocal kernel $\omega^\epsilon : \mathbb{R} \to \mathbb{R}$ satisfying $\omega^\epsilon \geq 0$ and $\int_0^\epsilon \omega^\epsilon(y)\,dy = 1$.

We denote (2.1) and (2.2) by the local and nonlocal models, respectively. The first natural question to ask is whether the nonlocal model (2.2) converges to the local model (2.1) as ϵ tends to zero. To answer the question, we consider the rescaled kernel $\omega^\epsilon(y) = \frac{1}{\epsilon}\omega(\frac{y}{\epsilon})$ such that it converges to Dirac point mass as ϵ tends to zero. Formally speaking, $V_\epsilon(\rho) \to v(\rho)$ as $\epsilon \to 0$; we recover (2.1) as ϵ tends to zero. But whether the corresponding solutions of (2.2) converge to a solution of (2.1), is the key question that has been studied in the literature; see [2–4, 8, 11]. Throughout this thesis, we denote the solutions of the local model (2.1) and nonlocal model (2.2) by u and ρ, respectively, and study the properties of these solutions.

2.1.1 Well-Posedness of the Local Model (2.1)

Since solutions of conservation laws may not be differentiable or even continuous, we cannot expect classical solutions. That is, we may not find a solution that is sufficiently differentiable and satisfies the PDE (2.1) for all points. Hence, we briefly mention the notions of weak solution and entropy solution.

Definition 2.1.1 (Weak solution of (2.1))**.** A function $u \in L^\infty(\mathbb{R} \times \mathbb{R}_+)$ is a weak solution of (2.1) with initial data $u_0 \in L^\infty(\mathbb{R})$ if the following identity holds for all test functions $\phi \in C^1(\mathbb{R} \times \mathbb{R}_+)$

$$\int_{\mathbb{R}_+} \int_{\mathbb{R}} u\phi_t + f(u)\phi_x \, dx \, dt + \int_{\mathbb{R}} u_0 \phi(x,0) \, dx = 0. \tag{2.4}$$

Since weak solutions are not necessarily unique, some additional conditions have to be imposed in order to pick a weak solution that describes the physical flow correctly. If the solutions contain discontinuities, then they must satisfy the following Rankine–Hugoniot condition:

$$s(t) = \frac{f(u_R) - f(u_L)}{u_R - u_L}, \tag{2.5}$$

where $s(t) = \gamma'(t)$ denotes the speed of a shock wave $\gamma(t)$, and u_L and u_R are continuously differentiable functions.

Theorem 2.1.2. *Let $\gamma \in C^1(\mathbb{R}_+)$ and $u \in L^\infty(\mathbb{R} \times \mathbb{R}_+)$ be on the form*

$$u(x,t) = \begin{cases} u_L & \text{if } x < \gamma(t) \\ u_R & \text{if } x > \gamma(t), \end{cases}$$

where u_L and u_R are continuously differentiable functions. Then u is a weak solution of the local model if and only if
(i) both u_L and u_R solve the local model in the classical sense.
(ii) $s(t)$ satisfy the Rankine–Hugoniot condition (2.5).

The above theorem says that if u contains two states separated by a shock wave that satisfies the Rankine–Hugoniot condition, then u is a weak solution of the local model. However, the Rankine–Hugoniot condition alone does not guarantee uniqueness of u. Therefore, we need to impose a condition that guarantees stability and uniqueness of weak solutions.

The well-posedness (i.e., there exists a unique solution) of the local model (2.1) is obtained through viscous approximation; see [7, Chapter 2]. The technique is to add a viscous term to the conservation law (2.1) in order to turn (2.1) from hyperbolic to parabolic. That is, we study the new parabolic PDE

$$u^\delta_t + f(u^\delta)_x = \delta u^\delta_{xx} \tag{2.6}$$

where $\delta > 0$. Formally, if we take $\delta \to 0$ we recover (2.1). Solutions of (2.6) possess similar properties as solutions of the heat equation, i.e., we obtain smooth solutions. Moreover, the solutions u^δ satisfy a useful property such as

$$\eta(u^\delta)_t + q(u^\delta)_x \leq 0 \tag{2.7}$$

where the entropy function $\eta : \mathbb{R} \to \mathbb{R}$ is any strictly convex function, and the entropy flux function $q : \mathbb{R} \to \mathbb{R}$ is defined as

$$q(u) := \int_0^u f'(s)\eta'(s) \, ds.$$

A limit $u = lim_{\delta\to 0} u^\delta$ is called a vanishing viscosity solution of the local model (2.1) if it satisfies (2.4), and passing $\delta \to 0$ in (2.7) gives us the following property:

$$\eta(u)_t + q(u)_x \leq 0. \tag{2.8}$$

The above property should be interpreted in the weak sense as usual, therefore the following entropy condition is used to select a unique weak solution of the local model:

$$\int_0^\infty \int_{\mathbb{R}} (\eta(u(x,t))\phi_t(x,t) + q(u(x,t))\phi_x(x,t))\, dx\, dt + \int_{\mathbb{R}} \eta(u_0(x))\phi(x,0)\, dx \geq 0 \tag{2.9}$$

for all test functions $\phi \in C_c^1(\mathbb{R} \times [0,\infty))$ with $\phi \geq 0$.

Definition 2.1.3 (Entropy solution)**.** A function $u \in L^\infty(\mathbb{R} \times \mathbb{R}_+)$ is an entropy solution of the local model (2.1) provided the following conditions:
(i) u is a weak solution of the local model (2.1)
(ii) u satisfies the entropy condition (2.9)

Particular choice for the entropy pairs are $\eta(u) = |u-k|$ with corresponding $q(u) = sign(u-k)(f(u)-f(k))$ for any $k \in \mathbb{R}$. Those pairs are called Kruzkhov entropy pairs. We say that a function satisfies the Kruzkov entropy condition if it satisfies

$$|u-k|_t + (sign(u-k)(f(u-f(k))))_x \leq 0. \tag{2.10}$$

The following Lemma helps us to determine whether a given function is an entropy solution of the local model (2.1).

Lemma 2.1.4. *A function $u \in L^\infty(\mathbb{R} \times \mathbb{R}_+)$ is an entropy solution of the local model* (2.1) *if and only if it satisfies the Kruzkov entropy condition* (2.10)*.*

The solutions u^δ of the viscous equation (2.6) satisfy the maximum principle, so $\|u^\delta\|_{L^\infty(\mathbb{R})} \leq \|u_0\|_{L^\infty(\mathbb{R})}$. Taking the limit $\delta \to 0$ gives us L^∞ bound $\|u\|_{L^\infty(\mathbb{R})} \leq \|u_0\|_{L^\infty(\mathbb{R})}$. On the other hand, the property (2.8) can be used to obtain $L^p(\mathbb{R})$ estimate on u such that $\|u\|_{L^p(\mathbb{R})} \leq \|u_0\|_{L^p(\mathbb{R})}$.

Let us now summarize the important properties of the entropy solution of the local model (2.1), which will play a crucial role in the stability analysis of the numerical methods in Chapter 4.

Theorem 2.1.5. *Assume that $f \in C^1(\mathbb{R})$ and that $u_0 \in L^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$. Then there exists a unique entropy solution u of* (2.1)*. Moreover, the following estimates hold:*

- *L^1 bound:*
$$\|u(\cdot,t)\|_{L^1(\mathbb{R})} \leq \|u_0\|_{L^1(\mathbb{R})}. \tag{2.11}$$

- *L^∞ bound:*
$$\|u(\cdot,t)\|_{L^\infty(\mathbb{R})} \leq \|u_0\|_{L^\infty(\mathbb{R})}. \tag{2.12}$$

- *TV bound: if $\|u_0\|_{TV} < \infty$, then*
$$\|u(\cdot,t)\|_{TV} \leq \|u_0\|_{TV}, \tag{2.13}$$

  where the TV-norm is defined by
$$\|g\|_{TV([a,b])} = \sup_{\mathcal{P}} \sum_{j=1}^{N-1} |g(x_{j+1}) - g(x_j)| \quad \textit{with } \mathcal{P} = \{a = x_1 < x_2 < \cdots < x_N = b\}.$$

- *Time continuity: if $\|u_0\|_{TV} < \infty$, then*

$$\|u(t) - u(s)\|_{L^1(\mathbb{R})} \leq |t - s|\, M\, \|U_0\|_{TV(\mathbb{R})}, \tag{2.14}$$

where

$$M = M(u_0) := \max_{u_- \leq u \leq u_+} |f'(u)|, \qquad u_- := \operatorname*{ess\,inf}_{x\in\mathbb{R}} u_0(x), \quad u_+ := \operatorname*{ess\,sup}_{x\in\mathbb{R}} u_0(x).$$

Furthermore, if u and v are the entropy solutions corresponding to initial data u_0 and v_0, respectively, then we have:

- *Monotonicity: if $u_0 \leq v_0$ for all $x \in \mathbb{R}$, then*

$$u(x,t) \leq v(x,t) \qquad \textit{for all } x \in \mathbb{R}. \tag{2.15}$$

2.1.2 Solutions to Riemann Problems

For numerical experiments we need entropy solutions for different initial data. For Riemann initial data, we have a solution formula which will be used in Section 4. The Riemann problem is the initial value problem

$$\begin{aligned} u_t + f(u)_x &= 0 \\ u(x,0) &= \begin{cases} u_L & \text{if } x < 0 \\ u_R & \text{if } x > 0. \end{cases} \end{aligned} \tag{2.16}$$

Solution of this equation is constructed as

$$u(x,t) = \begin{cases} u_L & \text{for } x \leq h'(u_L)t \\ (h')^{-1}(\frac{x}{t}) & \text{for } h'(u_L)t \leq x \leq h'(u_R)t \\ u_R & \text{for } x \geq h'(u_R)t, \end{cases} \tag{2.17}$$

where $h(u)$ is defined by

$$h(u) = \begin{cases} \sup\{g(u) \mid g \leq f \text{ and } g \text{ is convex on } [u_L, u_R]\} & \text{if } u_L < u_R, \\ \inf\{g(u) \mid g \geq f \text{ and } g \text{ is concave on } [u_R, u_L]\} & \text{if } u_R < u_L. \end{cases} \tag{2.18}$$

2.1.3 Well-Posedness of the Nonlocal Model (2.2)

We introduced above the concept of a weak solution, together with the corresponding entropy condition, for the local model (2.1). For the nonlocal model (2.2), analogous notions must also be considered. However, for our chosen flux $f(u) = u(1-u)$, the nonlocal model admits a unique weak solution ρ; see [1, 9]. In [1], the authors first derive a sequence of approximate solutions. They then establish $\mathbf{L}^\infty$, **BV**, and $\mathbf{L^1}$ estimates, as well as discrete entropy inequalities for this sequence. Using the Lax–Wendroff theorem, they conclude that the sequence converges to the unique weak entropy solution of the nonlocal model (2.2). Therefore, we only state the definition of a weak solution for the nonlocal model.

Definition 2.1.6 (Weak solution of (2.2))**.** A function $\rho \in L^\infty(\mathbb{R}\times\mathbb{R}_+)$ is a weak solution of (2.2) with initial data $\rho_0 \in L^\infty(\mathbb{R})$ if the following identity holds for all test functions $\phi \in C_c^1(\mathbb{R}\times\mathbb{R}_+)$:

$$\int_{\mathbb{R}_+}\int_{\mathbb{R}} (\rho\phi_t + V(\rho)\phi_x)\, dx\, dt + \int_{\mathbb{R}} \rho_0(x)\phi(x,0)\, dx = 0. \tag{2.19}$$

2.2 Questions Concerning the Local and Nonlocal Models

In Chapter 3, we develop numerical schemes for both models. Our aim is to analyze the stability of these schemes and to investigate the relationship between the corresponding solutions.

We use the following notation:

- u: the solution of the local model in the continuous setting.
- $u^{\Delta x}$: the numerical solution of the local model.
- ρ^{ϵ}: the solution of the nonlocal model in the continuous setting.
- $\rho^{\epsilon,\Delta x}$: the numerical solution of the nonlocal model.

This leads to the following natural questions:

- Does $u^{\Delta x}$ converge to u as $\Delta x \to 0$?
- Does ρ^{ϵ} converge to u as $\epsilon \to 0$?
- Does $\rho^{\epsilon,\Delta x}$ converge to u as both $\Delta x \to 0$ and $\epsilon \to 0$ simultaneously?
- Does $\rho^{\epsilon,\Delta x}$ converge to $u^{\Delta x}$ for fixed Δx as $\epsilon \to 0$?
- Does $\rho^{\epsilon,\Delta x}$ converge to ρ for fixed ϵ as $\Delta x \to 0$?

For clarity, we include the following diagram.

$$
\begin{tikzcd}
\rho^{\epsilon,\Delta x} \arrow[r, "\epsilon \to 0"] \arrow[d, "\Delta x \to 0"'] \arrow[dr, "{\epsilon \to 0,\ \Delta x \to 0}"] & u^{\Delta x} \arrow[d, "\Delta x \to 0"] \\
\rho^{\epsilon} \arrow[r, "\epsilon \to 0"] & u
\end{tikzcd}
$$

Figure 2.1: Diagram of limiting paths between the local and nonlocal models.

1. Regarding the convergence of $u^{\Delta x}$ to u, see [10, Chapter 4], where conservative, consistent, and monotone schemes converge to the entropy solution u of the local model (2.1).

2. The convergence of ρ^{ϵ} to the entropy solution u of the local model as $\epsilon \to 0$ has been studied in [2–4]. In [2], the authors consider the nonlocal exponential kernels $\omega^{\epsilon}(y) = \frac{1}{\epsilon} e^{-\frac{y}{\epsilon}}$, and obtained convergence of ρ^{ϵ} to u, under the condition that the initial data ρ_0 is uniformly positive. In [4], they obtained convergence results for a class of nonlocal kernels, but under one additional condition on ρ_0 that it satisfies a one-sided Lipschitz condition. To overcome the restrictions on the initial data ρ_0, the authors in [3] studied the stability analysis on the convolution term $(\rho * \omega^{\epsilon})$ which has better stability and convergence properties, and proved that the weak solution of the nonlocal model(2.2) converges to the entropy solution u of the local model (2.1).

3. Regarding asymptotic compatibility, i.e., the convergence of $\rho^{\epsilon,\Delta x}$ to u as both $\Delta x \to 0$ and $\epsilon \to 0$ simultaneously, see the recent studies in [8, 11]. We also provide in Chapter 4 the stability analysis used in [8].

Part II

First-order Numerical Methods

Chapter 3

First-order Numerical Methods

In this chapter, we derive numerical methods for solving both the local model (2.1) and nonlocal model (2.2). First, we derive a general approach for designing numerical methods for conservation laws. Then, we establish the numerical methods for our specific chosen flux $f(u) = u(1-u)$.

3.1 Finite Volume Scheme for a General flux f

Designing a finite difference method typically consists of four steps:

1. Discretizing the domain,
2. Satisfying the equation at discrete points,
3. Replacing derivatives by finite differences,
4. Solving the discretized problem.

However, step 3 requires that solutions are sufficiently smooth, but solutions of conservation laws may not be differentiable or even continuous. In addition, the unknown solution u of a conservation law may represent a density (as our model, u represents the average number of cars per m^2). For these reasons, we must adapt the steps above to design a suitable numerical method.

3.1.1 Discretizing the Domain and Control Volumes

For simplicity, we consider a uniform discretization of both the one-dimensional spatial domain $[x_L, x_R]$ and the temporal domain $[0, T]$. We split the spatial domain $\Omega = [x_L, x_R]$ into K smaller subdomains $\mathcal{C}_j$. Let

$$\Delta x = \frac{x_R - x_L}{K},$$

then we have

$$\Omega = \bigcup_{j=0}^{K-1} \mathcal{C}_j = \bigcup_{j=0}^{K-1} [x_{j-1/2},\, x_{j+1/2}) \tag{3.1}$$

where the left interface $x_{j-1/2}$ and the right interface $x_{j+1/2}$ of $\mathcal{C}_j$ are given by

$$x_{j-1/2} = x_j - \tfrac{\Delta x}{2}, \quad x_{j+1/2} = x_j + \tfrac{\Delta x}{2}.$$

Here x_j denotes the midpoint of each $\mathcal{C}_j$ given by

$$x_j = x_L + \Big(j + 1/2\Big)\Delta x, \quad j = 0, \ldots, K-1,$$

We have now split Ω into K smaller non-overlapping subdomains called control volumes $\mathcal{C}_j$. These control volumes are essential for the finite volume method, as pointwise evaluation does not make sense. Setting $\Delta t = \frac{T}{N}$, we also discretize the temporal domain $[0, T]$ as

$$0 = t^0 < t^1 < \cdots < t^N = T, \quad \text{where } t^n = n\Delta t,\ n = 0, \ldots, N.$$

3.1.2 Cell Averages and a Finite Volume Scheme

As solutions of conservation laws may be discontinuous, pointwise evaluation does not make sense. Instead, at each time level t^n, we take cell averages:

$$u_j^n \approx \frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} u(x, t^n)\, dx \tag{3.2}$$

Assuming now the cell averages are known at some time level t^n, we find the cell averages at next time level t^{n+1} as follows. Taking an integral of (2.1) over the domain $[x_{j-1/2}, x_{j+1/2}] \times [t^n, t^{n+1}]$ gives

$$\int_{t^n}^{t^{n+1}} \int_{x_{j-1/2}}^{x_{j+1/2}} u_t\, dx\, dt + \int_{t^n}^{t^{n+1}} \int_{x_{j-1/2}}^{x_{j+1/2}} f(u)_x\, dx\, dt = 0.$$

Applying the fundamental theorem of calculus yields

$$\begin{aligned} &\int_{x_{j-1/2}}^{x_{j+1/2}} u(x, t^{n+1})\, dx - \int_{x_{j-1/2}}^{x_{j+1/2}} u(x, t^n)\, dx \\ = -&\int_{t^n}^{t^{n+1}} f(u(x_{j+1/2}, t))\, dt + \int_{t^n}^{t^{n+1}} f(u(x_{j-1/2}, t))\, dt. \end{aligned} \tag{3.3}$$

Defining the numerical fluxes

$$\bar{F}_{j+\frac{1}{2}} = \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} f(u(x_{j+1/2}, t))\, dt \tag{3.4}$$

and dividing both sides of (10) by Δx, we obtain

$$u_j^{n+1} = u_j^n - \frac{\Delta t}{\Delta x}(\bar{F}_{j+1/2} - \bar{F}_{j-1/2}) \tag{3.5}$$

Note that the relation in (3.5) is not explicit, since $\bar{F}$ requires a priori knowledge of the exact solution. By introducing an appropriate approximation of the numerical flux $\bar{F}$ by F, we obtain a finite volume scheme for a general flux f:

$$u_j^{n+1} = u_j^n - \frac{\Delta t}{\Delta x}(F_{j+1/2} - F_{j-1/2}) \tag{3.6}$$

3.2 Lax–Friedrichs Scheme and Godunov scheme

We will now specify the numerical flux $F_{j+1/2}$ in (3.6) for both the local model (2.1) and nonlocal model (2.2).

3.2.1 Lax–Friedrichs Scheme and Godunov Scheme for the Local Model

Focusing on the local model, the finite volume scheme (3.6) can be written as

$$\begin{aligned} u_j^{n+1} &= H(u_{j-p}^n, \dots, u_{j+p}^n) \\ &= u_j^n - \frac{\Delta t}{\Delta x}\Big(F(u_{j-p+1}^n, \dots, u_{j+p}) - F(u_{j-p}, \dots, u_{j+p-1}^n)\Big) \end{aligned} \tag{3.7}$$

where H is an update function that depends on $2p+1$ points with $p \in \mathbb{N}$, and F is a numerical flux that depends on $2p$ points such that

$$F_{j+1/2} = F(u_{j-p+1}, \dots, u_{j+p}).$$

If we choose $p = 1$ we obtain a family of three-point finite volume schemes with corresponding two-point flux F as follows

$$u_j^{n+1} = H(u_{j-1}, u_j, u_{j+1}) = u_j^n - \frac{\Delta t}{\Delta x}\Big(F(u_j, u_{j+1}) - F(u_{j-1}, u_j)\Big). \tag{3.8}$$

The two-point flux function F can be specified, for instance, using Lax–Friedrichs, Godunov, Rusanov, and Engquist–Osher. But in this thesis, we focus on Lax–Friedrichs flux and Godunov flux. For our chosen flux $f(u) = u(1-u)$, we can specify the Lax–Friedrichs flux and Godunov flux:

$$\begin{aligned} F_{j+1/2} = F^{LxF}(u_j^n, u_{j+1}^n) &= \frac{1}{2}u_j^n(1-u_j^n) + \frac{1}{2}u_{j+1}^n(1-u_{j+1}^n) + \frac{\Delta x}{2\Delta t}(u_j^n - u_{j+1}^n) \\ &= \frac{1}{2}u_j^n(1-u_j^n) + \frac{1}{2}u_{j+1}^n(1-u_{j+1}^n) + \frac{\alpha}{2}(u_j^n - u_{j+1}^n) \end{aligned} \tag{3.9}$$

and

$$F_{j+1/2} = F^{God}(u_j^n, u_{j+1}^n) = u_j^n(1-u_{j+1}^n), \tag{3.10}$$

respectively. Here α is a positive numerical viscosity to be chosen later in the implementation.

3.2.2 Lax–Friedrichs Scheme and Godunov Scheme for the Nonlocal Model

To specify the numerical flux $F_{j+1/2}$ for the nonlocal model we need to approximate the integral term of V_ϵ given in (2.3). Let

$m := \lceil \frac{\epsilon}{\Delta x} \rceil$ denotes the number of cells involved in the nonlocal integral term (2.3),

and define

$$V_{j+1/2} := \int_0^\epsilon \omega^\epsilon(y)\rho(x_{j+1/2} + y, t)\, dy.$$

Assume the approximate cell averages ρ_j^n are known at time level t^n, following [1] and [8], we approximate $V_{j+1/2}$ in the way described below:

$$\begin{aligned} V_{j+1/2} &\approx \Delta x \sum_{k=0}^{m-1} \omega^\epsilon(k\Delta x)\rho(x_{j+1/2} + k\Delta x, t) \\ &= \Delta x \sum_{k=0}^{m-1} \omega^\epsilon(k\Delta x)\rho(x_{j+k+1/2}, t) \\ &\approx \Delta x \sum_{k=0}^{m-1} \omega^\epsilon(k\Delta x)\rho_{j+k}^n \qquad \text{by definition of cell averages} \\ &= \sum_{k=0}^{m-1} w_k \rho_{j+k}^n, \qquad w_k = \omega^\epsilon(k\Delta x)\Delta x. \end{aligned}$$

We denote $\{w_k\}_{k=0}^{m-1}$ as a set of numerical quadrature weights that are used to approximate the nonlocal kernel ω^ϵ. w_k can be given by

1. Left endpoint
$$w_k = \omega^\epsilon(k\Delta x)\Delta x \text{ for } k = 0, \ldots, m-1, \tag{3.11}$$

2. Normalized left endpoint
$$w_k = \frac{\omega^\epsilon(k\Delta x)\Delta x}{\sum_{k=0}^{m-1} \omega^\epsilon(k\Delta x)\Delta x} \text{ for } k = 0, \ldots, m-1, \tag{3.12}$$

3. Exact quadrature
$$w_k = \int_{k\Delta x}^{min\{(k+1)\Delta x, \epsilon\}} \omega^\epsilon(y)\, dy \text{ for } k = 0, \ldots, m-1. \tag{3.13}$$

Similar to the local model we consider here a family of finite volume schemes:

$$\begin{aligned} \rho_j^{n+1} &= \mathcal{H}(\rho_{j-1}^n, \rho_j^n, \rho_{j+1}^n \ldots, \rho_{j+m}^n) \\ &= \rho_j^n - \frac{\Delta t}{\Delta x}\Big(\mathcal{F}(\rho_j^n, \rho_{j+1}^n, q_j^n, q_{j+1}^n) - \mathcal{F}(\rho_{j-1}^n, \rho_j^n, q_{j-1}^n, q_j^n)\Big) \end{aligned} \tag{3.14}$$

where $\mathcal{H}$ is an update function depending on $m+2$ points, and $\mathcal{F}$ is a numerical flux depending on $m+1$ points with

$$q_j^n := \sum_{k=0}^{m-1} w_k\, \rho_{j+k}^n.$$

Now, the Lax–Friedrichs flux and Godunov flux can be specified as

$$\begin{aligned} F_{j+1/2} &= \mathcal{F}^{LxF}(\rho_j^n, \rho_{j+1}^n, q_j^n, q_{j+1}^n) \\ &= \frac{1}{2}\rho_j^n(1-q_j^n) + \frac{1}{2}\rho_{j+1}^n(1-q_{j+1}^n) + \frac{\alpha}{2}(\rho_j^n - \rho_{j+1}^n) \end{aligned} \tag{3.15}$$

and

$$F_{j+1/2} = \mathcal{F}^{God}(\rho_j^n, \rho_{j+1}^n, q_j^n, q_{j+1}^n) = \rho_j^n(1-q_{j+1}^n), \tag{3.16}$$

respectively.

3.2.3 Nonlocal-to-local Limit of Numerical Discretizing

In the previous subsections, we have developed a family of finite volume schemes for the local model and nonlocal model. In particular, we have developed a three-point update function H given in (3.8) for the local model. Does the update function $\mathcal{H}$ (3.14) for the nonlocal model converge to the three-point update function H for fixed Δx as $\epsilon \to 0$?

Proposition 3.2.1. *Assume the numerical quadrature weights w_k satisfy the following normalization condition:*

$$\sum_{k=0}^{m-1} w_k = 1.$$

Let $\Delta x > 0$. Then, when $\epsilon \le \Delta x$, the update function $\mathcal{H}$ given in (3.14) is equal to the three-point update function H given in (3.8).

Proof. It suffices to show that

$$q_j^n = \sum_{k=0}^{m-1} w_k\, \rho_{j+k}^n = \rho_j^n \qquad \text{whenever } \epsilon \le \Delta x.$$

Since then

$$\mathcal{F}(\rho_j^n, \rho_{j+1}^n, q_j^n, q_{j+1}^n) = \mathcal{F}(\rho_j^n, \rho_{j+1}^n, \rho_j^n, \rho_{j+1}^n),$$

which is a two-point numerical flux only depends on the local stencil values. That means $\mathcal{H}$ becomes a three-point update function for the local model.

If we have $\epsilon \le \Delta x$. Then, we have $m = 1$, i.e., only one cell of size Δx is involved in then nonlocal integral term (2.3). So, we have

$$q_j^n = \sum_{k=0}^{m-1} w_k\, \rho_{j+k}^n = w_0\, \rho_j^n,$$

and we must have $w_0 = 1$ by the normalization condition. This proves the claim. □

The proposition above tells us, if $\epsilon \le \Delta x$ and the numerical quadrature weights w_k satisfy the normalization condition, then the local method (3.8) and nonlocal method (3.14) are equal.

3.3 The Boundary Conditions and Stability

In this section, we consider two main aspects; the boundary conditions and stability. Usually, both the boundary conditions and stability require careful treatment when solving PDEs numerically.

3.3.1 The Boundary Conditions

We desire an algorithm for solving both local model (2.1) and nonlocal model (2.2) numerically. Using, for instance, the The Lax–Friedrichs scheme (3.15):

STEP 1. initialize ρ_j^0 by setting:

$$\rho_j^0 = \frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} \rho_0(x)\, dx, \quad \forall j \in \{0, \dots, K-1\}.$$

STEP 2. for $n = 0, \dots, N-1$ and $j \in \{0, \dots, K-1\}$, compute:

$$\begin{aligned}\rho_j^{n+1} &= \rho_j^n - \frac{\Delta t}{\Delta x}(F_{j+1/2} - F_{j-1/2}) \\ &= \rho_j^n - \frac{\Delta t}{\Delta x}\Big[\frac{1}{2}(\rho_{j+1}^n(1-q_{j+1}^n) - \rho_{j-1}^n(1-q_{j-1}^n)) - \frac{\alpha}{2}(\rho_{j+1}^n + \rho_{j-1}^n) + \alpha\, \rho_j^n\Big].\end{aligned}$$

The second step is also supposed to hold for all $j \in \{0, \dots, K-1\}$. However, for some nodes for example $j = 0$ and $j = K-1$ we obtain:

$$\rho_0^{n+1} = \rho_0^n - \frac{\Delta t}{\Delta x}\Big[\frac{1}{2}(\rho_1^n(1-q_1^n) - \rho_{-1}^n(1-q_{-1}^n)) - \frac{\alpha}{2}(\rho_1^n + \rho_{-1}^n) + \alpha\, \rho_0^n\Big]$$

and

$$\rho_{K-1}^{n+1} = \rho_{K-1}^n - \frac{\Delta t}{\Delta x}\Big[\frac{1}{2}(\rho_K^n(1-q_K^n) - \rho_{K-2}^n(1-q_{K-2}^n)) - \frac{\alpha}{2}(\rho_K^n + \rho_{K-2}^n) + \alpha\, \rho_{K-1}^n\Big],$$

respectively. We observe that for some nodes, the scheme involves undefined terms such as ρ_{-1}^n and $\rho_{K-1+\ell}^n$ for $\ell = 0, \dots m-1$. Therefore, without additional information the scheme is not valid for all nodes.

The above discussion indicate that specifying boundary conditions are needed. So, we take a look at how the boundary conditions are handled. There are several options for specifying the boundary conditions on the spatial domain $[x_L, x_R]$, but the choice depends on whether we have an initial value problem (IVP) or a boundary value problem (BVP).

- **Initial Value Problem (IVP):** Artificial boundary conditions are required to handle the truncated domain $[x_L, x_R]$. Truncation is necessarily for initial value problem on the entire real line $\mathbb{R}$ as computational domain needs to be bounded.
- **Boundary Value Problem (BVP):** Common choices include Periodic, Dirichlet, and Neumann boundary conditions.

In this thesis, we focus on **artificial boundary conditions** and **Periodic boundary conditions**. For a detailed discussion of the remaining types; see [10, p. 58]. **Artificial boundary conditions** and **Periodic boundary conditions** are implemented as

$$\rho_{-1}^n = \rho_0^n; \quad \rho_{K-1+\ell}^n = \rho_{K-1}^n \text{ for } \ell = 0, \dots m-1$$

and

$$\rho_{-1}^n = \rho_{K-1}^n; \quad \rho_{K-1+\ell}^n = \rho_\ell^n \text{ for } \ell = 0, \dots m-1,$$

respectively.

3.3.2 The Stability

We may face undesired oscillations when implementing the numerical schemes. The ratio, known as the CFL ratio,

$$\lambda = \frac{\Delta t}{\Delta x}$$

plays essential role in obtaining stable numerical schemes. Normally, some numerical schemes require stricter CFL condition than others. In [8], the stability analysis is done under the condition

$$\lambda \sum_{i=1}^{4} \left\| \frac{\partial g}{\partial \rho_i} \right\|_{L^\infty} < 1 \tag{3.17}$$

where g is a numerical flux.

For convenience, let us consider the Godunov numerical flux given by

$$g(\rho_L, \rho_R, q_L, q_R) = \rho_L(1 - q_R).$$

Computing the partial derivatives gives

$$\begin{aligned} \frac{\partial g}{\partial \rho_L} &= 1 - q_R, \qquad \frac{\partial g}{\partial \rho_R} = 0 \\ \frac{\partial g}{\partial q_L} &= 0, \qquad \frac{\partial g}{\partial q_R} = -\rho_L. \end{aligned} \tag{3.18}$$

Suppose that $0 \le \rho_L, \rho_R, q_L, q_R \le 1$, then we have

$$\begin{aligned} 1 &> \lambda \sum_{i=1}^{4} \left\| \frac{\partial g}{\partial \rho_i} \right\|_{L^\infty} \\ &= \lambda \Big(\left\| \frac{\partial g}{\partial \rho_L} \right\|_{L^\infty} + \left\| \frac{\partial g}{\partial \rho_R} \right\|_{L^\infty} + \left\| \frac{\partial g}{\partial q_L} \right\|_{L^\infty} + \left\| \frac{\partial g}{\partial q_R} \right\|_{L^\infty} \Big) \\ &= \lambda \Big(\left\| \frac{\partial g}{\partial \rho_L} \right\|_{L^\infty} + \left\| \frac{\partial g}{\partial q_R} \right\|_{L^\infty} \Big) \\ &= \lambda \Big(\sup_{q_R \in [0,1]} |1 - q_R| + \sup_{\rho_L \in [0,1]} |-\rho_L| \Big) \\ &= 2\lambda, \end{aligned} \tag{3.19}$$

A similar calculation can be carried out for the Lax–Friedrichs numerical flux. When the numerical viscosity parameter is chosen as $\alpha = 2$, the corresponding CFL condition becomes

$$\lambda < \frac{1}{4}.$$

Thus, among the schemes considered here, the Lax–Friedrichs scheme imposes the more restrictive stability condition. Therefore, in the numerical experiments in this thesis, we choose the CFL condition

$$\Delta t \leq \frac{1}{4}\Delta x \tag{3.20}$$

when implementing the numerical schemes.

Chapter 4

Asymptotic Compatibility (AC) of the First-Order Numerical Methods

A numerical scheme of the form (3.14) is called asymptotically compatible if its numerical solution converges to the unique entropy solution of the local model (2.1) as both $\Delta x \to 0$ and $\epsilon \to 0$ simultaneously; see [12]. In this chapter, we present the stability analysis of the numerical method, which provides the main ingredients needed to establish asymptotic compatibility.

4.1 Stability Analysis of the Numerical Methods

We consider the stability analysis of the numerical method given by (3.14). For the local model, if we have a conservative, consistent and monotone scheme, then the scheme converges to the correct solution; see [10, Chapter 4]. However, for the nonlocal model it is difficult to obtain a monotone scheme. Nevertheless, under certain assumptions on the initial data, Huang and Du; see [8], have deduced a monotone numerical scheme. Further, they have proved the asymptotic compatibility of a class of finite-volume schemes for the nonlocal traffic-flow model. To make the analysis easier to follow, we restrict ourselves to the Godunov-type flux and present the corresponding stability analysis in this chapter.

We start making the necessary assumptions on the nonlocal kernel $\omega^\epsilon(y)$, initial data ρ_0, numerical quadrature weights w_k and CFL condition λ.

Assumption 4.1.1. The nonlocal kernel $\omega^\epsilon : \mathbb{R} \to \mathbb{R}$ satisfying $\omega^\epsilon \geq 0$ and $\int_0^\epsilon \omega^\epsilon(y)\, dy = 1$ is given by

$$\omega^\epsilon(y) = \begin{cases} \frac{1}{\epsilon}\omega\left(\frac{y}{\epsilon}\right), & y \in [0, \epsilon] \\ 0 & \text{otherwise} \end{cases}$$

where $\omega \in C_c^1([0,1])$ is a nonnegative, strictly decreasing probability density satisfying

$$\int_0^1 \omega(z)\, dz = 1.$$

Assumption 4.1.2. The initial data ρ_0 belongs to $L^\infty(\mathbb{R})$, has bounded total variation, and satisfies

$$0 \leq \rho_{\min} \leq \rho_0(x) \leq \rho_{\max} \leq 1, \qquad \text{for all } x \in \mathbb{R}.$$

Assumption 4.1.3. The numerical quadrature weights $\{w_k\}_{0\leq k \leq m-1}$ satisfy

1. $$\omega^\epsilon(k\Delta x)\Delta x \geq w_k \geq \omega^\epsilon((k+1)\Delta x)\Delta x \qquad \text{for } k = 0, 1, \ldots, m-1$$

2.
$$w_{k-1} - w_k \geq c\, m^{-2}, \qquad \text{for } k = 1, 2, \dots, m,$$
with the convention $w_m = 0$, and for some constant $c > 0$ depending only on the kernel ω.

3. normalization condition
$$\sum_{k=0}^{m-1} w_k = 1.$$

Assumption 4.1.4. The CFL condition λ satisfies
$$\lambda < \frac{1}{2}.$$

4.1.1 Maximum Principle

It is easy to establish the maximum principle if we have monotone numerical schemes. So, we check first if the update function $\mathcal{H}$ with corresponding Godunov numerical flux g is monotone. To do that, suppose Assumptions 4.1.1–4.1.4 holds. In addition, suppose that
$$0 \leq \rho_{\min} \leq \rho_{j+k}^n \leq \rho_{\max} \leq 1 \qquad \text{for } k = -1, 0, 1, \dots, m.$$

Then we have
$$q_j^n = \sum_{k=0}^{m-1} w_k \rho_{j+k}^n \leq \sum_{k=0}^{m-1} w_k \rho_{\max} = \rho_{\max}, \tag{4.1}$$
where we have used $\sum_{k=0}^{m-1} w_k = 1$ by the normalization condition. By a similar argument q_j^n is lower bounded by $\rho_{\min}$. Consequently, we have
$$0 \leq q_j^n \leq 1 \text{ for all } j \in \mathbb{Z}, n \in \mathbb{N}. \tag{4.2}$$

The update function $\mathcal{H}$ with respect to the Godunov numerical flux
$$g(\rho_L, \rho_R, q_L, q_R) = \rho_L(1 - q_R)$$
is given by
$$\mathcal{H}(\rho_{j-1}^n, \rho_j^n, \rho_{j+1}^n, \dots, \rho_{j+m}^n) = \rho_j^n + \lambda \left[\rho_{j-1}^n(1 - q_j^n) - \rho_j^n(1 - q_{j+1}^n)\right]. \tag{4.3}$$

Taking the partial derivatives gives
$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial \rho_{j-1}^n} &= \lambda(1 - q_j^n)\\
\frac{\partial \mathcal{H}}{\partial \rho_j^n} &= 1 - \lambda\left[(1 - q_{j+1}^n) + w_0 \rho_{j-1}^n\right]\\
\frac{\partial \mathcal{H}}{\partial \rho_{j+k}^n} &= \lambda\left(w_{k-1}\rho_j^n - w_k \rho_{j-1}^n\right) \qquad \text{for } k = 1, 2, \dots, m, \text{with the convention } w_m = 0.
\end{aligned}$$

$\mathcal{H}$ is nondecreasing with respect to ρ_{j-1}^n, because
$$\frac{\partial \mathcal{H}}{\partial \rho_{j-1}^n} = \lambda(1 - q_j^n) \geq 0 \qquad \text{since } 0 \leq q_j^n \leq 1 \text{ due to (4.2)}. \tag{4.4}$$

Next, we check if $\mathcal{H}$ is nondecreasing with respect to ρ_j^n. We desire to have that
$$\frac{\partial \mathcal{H}}{\partial \rho_j^n} = 1 - \lambda\left[(1 - q_{j+1}^n) + w_0 \rho_{j-1}^n\right] \geq 0.$$

Since $\lambda < \frac{1}{2}$ by 4.1.4, it suffices to show that

$$(1 - q_{j+1}^n) + w_0 \rho_{j-1}^n \leq 2.$$

So

$$\begin{aligned}(1 - q_{j+1}^n) + w_0 \rho_{j-1}^n &\leq (1 - q_{j+1}^n) + \rho_{j-1}^n \qquad \text{since } 0 \leq w_0 \leq 1 \text{ and } \rho_{j-1}^n \geq 0 \\ &\leq 1 + 1 = 2 \qquad \text{since } 0 \leq q_{j+1}^n \leq 1 \text{ and } 0 \leq \rho_{j-1}^n \leq 1\end{aligned}$$

Therefore, $\mathcal{H}$ is nondecreasing with respect to ρ_j^n.

For $k = m$, the convention $w_m = 0$ gives

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial \rho_{j+m}^n} &= \lambda \left(w_{k-1} \rho_j^n - w_k \rho_{j-1}^n \right) \\ &= \lambda (w_{m-1} \rho_j^n) \\ &\geq \lambda c m^{-2} \rho_j^n \qquad \text{by Assumption 4.1.3} \\ &\geq 0\end{aligned}$$

Hence, $\mathcal{H}$ is nondecreasing with respect to ρ_{j+m}^n We consider now for $k = 1, 2, \ldots, m-1$. The derivatives are given by

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial \rho_{j+k}^n} &= \lambda \left(w_{k-1} \rho_j^n - w_k \rho_{j-1}^n \right) \\ &= \lambda \Big[(w_{k-1} - w_k) \rho_j^n + w_k (\rho_j^n - \rho_{j-1}^n) \Big].\end{aligned}$$

Both the terms ρ_j^n and w_k are nonnegative, and by Assumption 4.1.3 we have

$$(w_{k-1} - w_k) \geq c m^{-2} > 0.$$

We assume first that $(\rho_j^n - \rho_{j-1}^n) \geq 0$, then the term

$$\lambda \Big[(w_{k-1} - w_k) \rho_j^n + w_k (\rho_j^n - \rho_{j-1}^n) \Big]$$

is nonnegative.

We assume now that $(\rho_j^n - \rho_{j-1}^n) < 0$, then the first term

$$(w_{k-1} - w_k) \rho_j^n$$

inside the brackets is still nonnegative. However, the second term

$$w_k (\rho_j^n - \rho_{j-1}^n)$$

inside the brackets is negative. Consequently, we may have $\frac{\partial \mathcal{H}}{\partial \rho_{j+k}^n} < 0$.

Thus, $\mathcal{H}$ is nondecreasing only with respect to ρ_{j-1}^n, ρ_j^n and ρ_{j+m}^n. So, we can not conclude that $\mathcal{H}$ is nondecreasing in each of its arguments based only on Assumptions 4.1.1–4.1.4. Therefore, one can not prove the maximum principle by showing $\mathcal{H}$ is monotone. Instead, we prove the maximum principle by applying the mean value theorem.

Theorem 4.1.5 (Maximum principle)**.** *Suppose Assumptions 4.1.1–4.1.4 holds. Let $\{\rho_j^n\}_{j \in \mathbb{Z}}^{n \in \mathbb{N}}$ be the numerical solution computed by (4.3). Then, the uniform bound holds:*

$$\rho_{\min} \leq \rho_j^n \leq \rho_{\max} \qquad \textit{for all } j \in \mathbb{Z}, n \in \mathbb{N}$$

where

$$\rho_{\min} := \inf_{x \in \mathbb{R}} \rho_0 \geq 0 \quad \textit{and } \rho_{\max} := \sup_{x \in \mathbb{R}} \rho_0 \leq 1.$$

The proof is done by induction.

Proof. For $n = 0$, it is obvious that $\rho_{\min} \leq \rho_j^0 \leq \rho_{\max}$ for all $j \in \mathbb{Z}$. Assume now that the result holds for some $n \in \mathbb{N}$, that is,

$$\rho_{\min} \leq \rho_j^n \leq \rho_{\max} \qquad \text{for all } j \in \mathbb{Z}.$$

We show that it also holds for $n+1$. We must show that

$$\rho_{\min} \leq \rho_j^{n+1} \leq \rho_{\max} \qquad \text{for all } j \in \mathbb{Z},$$

or equivalently

$$\rho_{\min} \leq \mathcal{H}(\rho_{j-1}^n, \rho_j^n, \rho_{j+1}^n, \dots, \rho_{j+m}^n) \leq \rho_{\max} \qquad \text{for all } j \in \mathbb{Z}.$$

Since

$$\mathcal{H}(\rho_{\min}, \rho_{\min}, \rho_{\min}, \dots, \rho_{\min}) = \rho_{\min},$$

we write

$$\mathcal{H}(\rho_{j-1}^n, \rho_j^n, \rho_{j+1}^n, \dots, \rho_{j+m}^n) - \rho_{\min} = \Delta\mathcal{H}_1 + \Delta\mathcal{H}_2,$$

where

$$\Delta\mathcal{H}_1 = \mathcal{H}(\rho_{j-1}^n, \rho_j^n, \rho_{j+1}^n, \dots, \rho_{j+m}^n) - \mathcal{H}(\rho_{\min}, \rho_{\min}, \rho_{j+1}^n, \dots, \rho_{j+m}^n)$$

and

$$\Delta\mathcal{H}_2 = \mathcal{H}(\rho_{\min}, \rho_{\min}, \rho_{j+1}^n, \dots, \rho_{j+m}^n) - \mathcal{H}(\rho_{\min}, \rho_{\min}, \rho_{\min}, \dots, \rho_{\min}).$$

Focusing on $\Delta\mathcal{H}_1$, by the mean value theorem there exists a point $(\tilde{\rho}_{j-1}^n, \tilde{\rho}_j^n)$ on the line segment from $(\rho_{\min}, \rho_{\min})$ to (ρ_{j-1}^n, ρ_j^n) such that

$$\begin{aligned}
\Delta\mathcal{H}_1 &= \mathcal{H}(\rho_{j-1}^n, \rho_j^n, \rho_{j+1}^n, \dots, \rho_{j+m}^n) - \mathcal{H}(\rho_{\min}, \rho_{\min}, \rho_{j+1}^n, \dots, \rho_{j+m}^n) \\
&= \frac{\partial\mathcal{H}}{\partial\rho_{j-1}^n}(\tilde{\rho}_{j-1}^n, \tilde{\rho}_j^n, \rho_{j+1}^n, \dots, \rho_{j+m}^n)(\rho_{j-1}^n - \rho_{\min}) + \frac{\partial\mathcal{H}}{\partial\rho_j^n}(\tilde{\rho}_{j-1}^n, \tilde{\rho}_j^n, \rho_{j+1}^n, \dots, \rho_{j+m}^n)(\rho_j^n - \rho_{\min}),
\end{aligned}$$

with

$$0 \leq \rho_{\min} \leq \tilde{\rho}_{j-1}^n \leq \rho_{\max} \text{ and } 0 \leq \rho_{\min} \leq \tilde{\rho}_j^n \leq \rho_{\max}.$$

We want to show that $\Delta\mathcal{H}_1 \geq 0$. It is enough to show that the two terms

$$\frac{\partial\mathcal{H}}{\partial\rho_{j-1}^n}(\tilde{\rho}_{j-1}^n, \tilde{\rho}_j^n, \rho_{j+1}^n, \dots, \rho_{j+m}^n)(\rho_{j-1}^n - \rho_{\min}) \tag{4.5}$$

and

$$\frac{\partial\mathcal{H}}{\partial\rho_j^n}(\tilde{\rho}_{j-1}^n, \tilde{\rho}_j^n, \rho_{j+1}^n, \dots, \rho_{j+m}^n)(\rho_j^n - \rho_{\min}) \tag{4.6}$$

are nonnegative. Since we have that

$$(\rho_{j-1}^n - \rho_{\min}) \geq 0, \quad \text{and } (\rho_j^n - \rho_{\min}) \geq 0,$$

it suffices to show that

$$\frac{\partial\mathcal{H}}{\partial\rho_{j-1}^n}(\tilde{\rho}_{j-1}^n, \tilde{\rho}_j^n, \rho_{j+1}^n, \dots, \rho_{j+m}^n) \geq 0 \quad \text{and } \frac{\partial\mathcal{H}}{\partial\rho_j^n}(\tilde{\rho}_{j-1}^n, \tilde{\rho}_j^n, \rho_{j+1}^n, \dots, \rho_{j+m}^n) \geq 0.$$

Since the derivative of $\mathcal{H}$ with respect to ρ_{j-1}^n is given by

$$\frac{\partial\mathcal{H}}{\partial\rho_{j-1}^n}(\rho_{j-1}^n, \rho_j^n, \rho_{j+1}^n, \dots, \rho_{j+m}^n) = \lambda(1 - q_j^n),$$

we obtain

$$\frac{\partial \mathcal{H}}{\partial \rho_{j-1}^n}(\tilde{\rho}_{j-1}^n, \tilde{\rho}_j^n, \rho_{j+1}^n, \ldots, \rho_{j+m}^n) = \lambda(1 - \tilde{q}_j^n),$$

where

$$\tilde{q}_j^n := w_0 \tilde{\rho}_j^n + \sum_{k=1}^{m-1} w_k \rho_{j+k}^n.$$

By the induction hypothesis, we have

$$0 \le \rho_{\min} \le \rho_{j+k}^n \le \rho_{\max} \le 1 \qquad \text{for } k = -1, 0, 1, \ldots, m.$$

So, by a similar argument as (4.1), we have $0 \le \tilde{q}_j^n \le 1$. Therefore, we obtain

$$\frac{\partial \mathcal{H}}{\partial \rho_{j-1}^n}(\tilde{\rho}_{j-1}^n, \tilde{\rho}_j^n, \rho_{j+1}^n, \ldots, \rho_{j+m}^n) = \lambda(1 - \tilde{q}_j^n) \ge 0.$$

So, (4.5) is nonnegative.

It remains to show that (4.6) is nonnegative. The derivative of $\mathcal{H}$ with respect to ρ_j^n is given by

$$\frac{\partial \mathcal{H}}{\partial \rho_j^n}(\rho_{j-1}^n, \rho_j^n, \rho_{j+1}^n, \ldots, \rho_{j+m}^n) = 1 - \lambda\left[(1 - q_{j+1}^n) + w_0 \rho_{j-1}^n\right].$$

Then, we have

$$\frac{\partial \mathcal{H}}{\partial \rho_j^n}(\tilde{\rho}_{j-1}^n, \tilde{\rho}_j^n, \rho_{j+1}^n, \ldots, \rho_{j+m}^n) = 1 - \lambda\left[(1 - q_{j+1}^n) + w_0 \tilde{\rho}_{j-1}^n\right].$$

Since $\lambda < \frac{1}{2}$ by Assumption 4.1.4, and since $0 \le q_{j+1}^n \le 1$ and $0 \le w_0 \tilde{\rho}_{j-1}^n \le 1$, we obtain

$$1 - \lambda\left[(1 - q_{j+1}^n) + w_0 \tilde{\rho}_{j-1}^n\right] \ge 0.$$

Thus, (4.6) is nonnegative as well, and we conclude that $\Delta\mathcal{H}_1 \ge 0$.

Next we show that $\Delta\mathcal{H}_2 \ge 0$. By the mean value theorem, there exist intermediate values

$$0 \le \rho_{\min} \le \tilde{\rho}_{j+k}^n \le \rho_{\max} \le 1 \qquad \text{for } k = 1, 2, \ldots, m,$$

such that

$$\begin{aligned}\Delta\mathcal{H}_2 &= \mathcal{H}(\rho_{\min}, \rho_{\min}, \rho_{j+1}^n, \ldots, \rho_{j+m}^n) - \mathcal{H}(\rho_{\min}, \rho_{\min}, \rho_{\min}, \ldots, \rho_{\min}) \\ &= \sum_{k=1}^{m} \frac{\partial \mathcal{H}}{\partial \rho_{j+k}^n}(\rho_{\min}, \rho_{\min}, \tilde{\rho}_{j+1}^n, \ldots, \tilde{\rho}_{j+m}^n)(\rho_{j+k}^n - \rho_{\min}).\end{aligned}$$

In order to have $\Delta\mathcal{H}_2 \ge 0$, we must show that

$$\frac{\partial \mathcal{H}}{\partial \rho_{j+k}^n}(\rho_{\min}, \rho_{\min}, \tilde{\rho}_{j+1}^n, \ldots, \tilde{\rho}_{j+m}^n)(\rho_{j+k}^n - \rho_{\min}) \ge 0 \qquad \text{for } k = 1, 2, \ldots, m.$$

Since

$$(\rho_{j+k}^n - \rho_{\min}) \ge 0 \qquad \text{for } k = 1, 2, \ldots, m,$$

it suffices to show that

$$\frac{\partial \mathcal{H}}{\partial \rho_{j+k}^n}(\rho_{\min}, \rho_{\min}, \tilde{\rho}_{j+1}^n, \ldots, \tilde{\rho}_{j+m}^n) \ge 0 \qquad \text{for } k = 1, 2, \ldots, m.$$

For $k = m$, using the convention $w_m = 0$, the derivative of $\mathcal{H}$ with respect to ρ_{j+m}^n is given by

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial \rho_{j+m}^n}(\rho_{j-1}^n, \rho_j^n, \rho_{j+1}^n, \ldots, \rho_{j+m}^n) &= \lambda\left(w_{m-1}\rho_j^n - w_m\rho_{j-1}^n\right) \\ &= \lambda w_{m-1}\rho_j^n\end{aligned}$$

Therefore,

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial \rho_{j+m}^n}(\rho_{\min}, \rho_{\min}, \tilde{\rho}_{j+1}^n, \ldots, \tilde{\rho}_{j+m}^n) &= \lambda w_{m-1}\rho_{\min} \\ &\geq \lambda c m^{-2}\rho_{\min} \qquad \text{by Assumption 4.1.3} \\ &\geq 0.\end{aligned}$$

For $k = 1, \ldots, m-1$, since we have

$$\frac{\partial \mathcal{H}}{\partial \rho_{j+k}^n}(\rho_{j-1}^n, \rho_j^n, \rho_{j+1}^n, \ldots, \rho_{j+m}^n) = \lambda\left(w_{k-1}\rho_j^n - w_k\rho_{j-1}^n\right),$$

we obtain

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial \rho_{j+k}^n}(\rho_{\min}, \rho_{\min}, \tilde{\rho}_{j+1}^n, \ldots, \tilde{\rho}_{j+m}^n) &= \lambda\left(w_{k-1}\rho_{\min} - w_k\rho_{\min}\right) \\ &= \lambda\rho_{\min}(w_{k-1} - w_k)\end{aligned}$$

By Assumption 4.1.3, we have $(w_{k-1} - w_k) \geq cm^{-2} > 0$. Therefore,

$$\frac{\partial \mathcal{H}}{\partial \rho_{j+k}^n}(\rho_{\min}, \rho_{\min}, \tilde{\rho}_{j+1}^n, \ldots, \tilde{\rho}_{j+m}^n) = \lambda\rho_{\min}\left(w_{k-1} - w_k\right) \geq 0 \qquad \text{for } k = 1, \ldots, m-1.$$

Hence $\Delta\mathcal{H}_2 \geq 0$. We have shown that both $\Delta\mathcal{H}_1 \geq 0$ and $\Delta\mathcal{H}_2 \geq 0$. Which implies

$$\mathcal{H}(\rho_{j-1}^n, \rho_j^n, \rho_{j+1}^n, \ldots, \rho_{j+m}^n) - \rho_{\min} = \Delta\mathcal{H}_1 + \Delta\mathcal{H}_2 \geq 0,$$

and therefore

$$\mathcal{H}(\rho_{j-1}^n, \rho_j^n, \rho_{j+1}^n, \ldots, \rho_{j+m}^n) \geq \rho_{\min}.$$

The upper bound can be shown by similar argument. Thus, by induction,

$$\rho_{\min} \leq \rho_j^n \leq \rho_{\max} \qquad \text{for all } j \in \mathbb{Z} \quad \text{and } n \in \mathbb{N}.$$

□

4.1.2 One-sided Lipschitz Estimate

Next, we prove the one-sided Lipschitz condition. The one-sided Lipschitz condition will play a crucial role in showing the monotonicity of the update function $\mathcal{H}$ given in (4.3).

Lemma 4.1.6. *Suppose Assumptions 4.1.1, 4.1.3 and 4.1.4 hold. In addition, suppose that*

1. *the initial data ρ_0 belongs to $L^\infty(\mathbb{R})$, has bounded total variation, and satisfies*

$$0 < \rho_{\min} \leq \rho_0(x) \leq 1, \qquad \text{for all } x \in \mathbb{R}$$

and

$$-\frac{\rho_0(y) - \rho_0(x)}{y - x} \leq L \qquad \text{for all } x \neq y \in \mathbb{R}$$

for some constant $L > 0$.

2.

$$c\rho_{\min} \geq 2L\omega(0)\epsilon, \qquad \textit{where } c \textit{ is as in Assumption 4.1.3.}$$

3.

$$\epsilon = m\Delta x \qquad \textit{where } m \geq 1 \textit{ donets the number of cells.}$$

Let $\{\rho_j^n\}_{j\in\mathbb{Z}}^{n\in\mathbb{N}}$ *be the numerical solution computed by* (4.3)*. Then, the numerical differences*

$$r_j^n = \rho_{j+1}^n - \rho_j^n, \qquad j \in \mathbb{Z}, n \in \mathbb{N},$$

satisfy the one-sided Lipschitz condition

$$r_j^n \geq -L\Delta x, \qquad j \in \mathbb{Z}, n \in \mathbb{N}. \tag{4.7}$$

We highlight the main results in the proof rather rigorously. We begin by considering the structure of

$$r_j^{n+1} = \rho_{j+1}^{n+1} - \rho_j^{n+1}.$$

By definition in (3.14), we have

$$\rho_{j+1}^{n+1} = \rho_{j+1}^n + \lambda\Big[g(\rho_j^n, \rho_{j+1}^n, q_j^n, q_{j+1}^n) - g(\rho_{j+1}^n, \rho_{j+2}^n, q_{j+1}^n, q_{j+2}^n)\Big]$$

and

$$\rho_j^{n+1} = \rho_j^n + \lambda\Big[g(\rho_{j-1}^n, \rho_j^n, q_{j-1}^n, q_j^n) - g(\rho_j^n, \rho_{j+1}^n, q_j^n, q_{j+1}^n)\Big],$$

where g is the Godunov numerical flux. Taking the difference gives us

$$\begin{aligned} r_j^{n+1} &= \rho_{j+1}^{n+1} - \rho_j^{n+1} \\ &= (\rho_{j+1}^n - \rho_j^n) + \lambda\Big[2g(\rho_j^n, \rho_{j+1}^n, q_j^n, q_{j+1}^n) - g(\rho_{j-1}^n, \rho_j^n, q_{j-1}^n, q_j^n) - g(\rho_{j+1}^n, \rho_{j+2}^n, q_{j+1}^n, q_{j+2}^n)\Big] \\ &= r_j^n + \lambda\Big[2g(\rho_j^n, \rho_{j+1}^n, q_j^n, q_{j+1}^n) - g(\rho_{j-1}^n, \rho_j^n, q_{j-1}^n, q_j^n) - g(\rho_{j+1}^n, \rho_{j+2}^n, q_{j+1}^n, q_{j+2}^n)\Big]. \end{aligned}$$

As g is quadratic, r_j^{n+1} can be written as

$$r_j^{n+1} = \sum_{k=-1}^{m} c_{j,k}^n r_{j+k}^n + 2\lambda(r_j^n)^2, \tag{4.8}$$

where

$$\begin{aligned} c_{j,-1}^n &= \lambda(1 - q_j^n) \\ c_{j,0}^n &= 1 - \lambda(1 - q_j^n) + \lambda p_{j,0}^n - 2\lambda r_j^n \\ c_{j,k}^n &= \lambda p_{j,k}^n, \qquad \text{for } k = 1, 2, \ldots, m, \end{aligned}$$

and

$$p_{j,k}^n = w_{k-1}\rho_{j+1}^n - w_k\rho_j^n + w_k r_j^n \qquad \text{with convention } w_{-1} = w_m = 0.$$

Now we prove the Lemma by induction

Proof of Lemma 4.1.6. Following [8], we do the analysis for $\Delta x = \epsilon m^{-1}$. The one-sided Lipschitz condition on ρ_0 gives $r_j^0 \geq -L\Delta x$ for all $j \in \mathbb{Z}$. Assume now that (4.7) holds for some $n \in \mathbb{N}$, that is,

$$r_j^n \geq -L\Delta x, \qquad j \in \mathbb{Z}.$$

We show also (4.7) holds for $n+1$. It suffices to show that r_j^{n+1} is a convex combination of $r_{j-1}^n, r_j^n, \ldots, r_{j+m}^n$ plus the nonnegative term $2\lambda(r_j^n)^2$, since then

$$\inf_{j\in\mathbb{Z}} r_j^{n+1} \geq \inf_{j\in\mathbb{Z}} r_j^n \geq -L\Delta x,$$

which gives the desired result.

Since the term $2\lambda(r_j^n)^2$ in (4.8) is nonnegative and $\sum_{k=-1}^m c_{j,k}^n = 1$, it remains to show $c_{j,k}^n \geq 0$ for all $k = -1, 0, 1, 2, \ldots, m$.

Clearly, $c_{j,-1}^n \geq 0$ because we have shown that in (4.4)

$$c_{j,-1}^n = \frac{\partial \mathcal{H}}{\partial \rho_{j-1}^n} = \lambda(1-q_j^n) \geq 0 \qquad \text{as } 0 \leq q_j^n \leq 1.$$

For $k = 0$, we have

$$\begin{aligned} c_{j,0}^n &= 1 - \lambda(1-q_j^n) + \lambda p_{j,0}^n - 2\lambda r_j^n \\ &= 1 - \lambda(1-q_j^n) + \lambda\Big(-w_0\rho_j^n + w_0 r_j^n\Big) - 2\lambda r_j^n \\ &= 1 - \lambda\Big[(1-q_j^n) + w_0\rho_j^n + (2-w_0)r_j^n\Big] \end{aligned}$$

In order to have $c_{j,0}^n \geq 0$, it holds to show that $\lambda\Big[(1-q_j^n) + w_0\rho_j^n + (2-w_0)r_j^n\Big] \leq 1$. But by (3.19) we have $\lambda < \frac{1}{2}$. Therefore, it is sufficient to show that

$$(1-q_j^n) + w_0\rho_j^n + (2-w_0)r_j^n \leq 2.$$

Note that all the terms $(1-q_j^n)$, $w_0\rho_j^n$ and $(2-w_0)$ are nonnegative since we have $0 \leq q_j^n \leq 1$, $0 \leq w_0 \leq 1$ and $0 \leq \rho_j^n \leq 1$. We assume first that $r_j^n \leq 0$, then we have $(2-w_0)r_j^n \leq 0$ and

$$\begin{aligned} (1-q_j^n) + w_0\rho_j^n + (2-w_0)r_j^n &\leq (1-q_j^n) + w_0\rho_j^n \\ &\leq (1-q_j^n) + \rho_j^n \\ &\leq (1-q_j^n) + \rho_{\max} \qquad \text{By maximum principle Theorem 4.1.5} \\ &\leq 2 \end{aligned}$$

If $r_j^n > 0$, then we have $(2-w_0)r_j^n > 0$. In this case, the term

$$(1-q_j^n) + w_0\rho_j^n + (2-w_0)r_j^n$$

is nonnegative and we must show that

$$(1-q_j^n) + w_0\rho_j^n + (2-w_0)r_j^n \leq 2,$$

or equivalently

$$(1-q_j^n) + w_0\rho_j^n \leq 2 - (2-w_0)r_j^n \tag{4.9}$$

Huang and Du [8, p. 13], claimed that this follows from Assumptions 4.1.1–4.1.4 and the conditions in Lemma 4.1.6. But this does not seem as obvious as they claim. Because from the assumptions we can estimate the right hand side of (4.9) as

$$\begin{aligned} (1-q_j^n) + w_0\rho_j^n &\leq (1-q_j^n) + \rho_j^n \\ &\leq (1-q_j^n) + \rho_{\max} \qquad \text{By maximum principle Theorem 4.1.5} \\ &\leq 2, \end{aligned}$$

and the right hand side of (4.9) is strictly less than 2, since the term $(2 - w_0)r_j^n$ positive. Since we have not got enough time to investigate this part properly, we continue to prove the rest of the proof.

For $k = 1, 2, \ldots, m$, we have $c_{j,k}^n = \lambda p_{j,k}^n$. So, it suffices to show that $p_{j,k}^n \geq 0$ for $k = 1, 2, \ldots, m$. We have

$$\begin{aligned} p_{j,k}^n &= w_{k-1}\rho_{j+1}^n - w_k\rho_j^n + w_k r_j^n \\ &= (w_{k-1} - w_k)\rho_{j+1}^n + w_k(\rho_{j+1}^n - \rho_j^n) + w_k r_j^n \\ &= (w_{k-1} - w_k)\rho_{j+1}^n + w_k r_j^n + w_k r_j^n \\ &= (w_{k-1} - w_k)\rho_{j+1}^n + 2w_k r_j^n. \end{aligned}$$

The terms $(w_{k-1} - w_k)$, ρ_{j+1}^n and $2w_k$ are nonnegative since we have

$$(w_{k-1} - w_k) \geq cm^{-2}, \quad w_k \geq 0 \qquad \text{by Assumptions 4.1.2–4.1.3},$$

and

$$\rho_{j+1}^n \geq \rho_{\min} > 0 \qquad \text{by maximum principle, Theorem 4.1.5.}$$

So, if $r_j^n \geq 0$, then we have $2w_k r_j^n \geq 0$ and

$$\begin{aligned} (w_{k-1} - w_k)\rho_{j+1}^n + 2w_k r_j^n &\geq cm^{-2}\rho_{\min} + 2w_k r_j^n \\ &\geq cm^{-2}\rho_{\min} \geq 0 \end{aligned}$$

If $r_j^n < 0$, then we have

$$\begin{aligned} w_k &\leq \omega^{\epsilon}(k\Delta x)\Delta x && \text{by Assumption 4.1.3} \\ &= \frac{1}{\epsilon}\omega\left(\frac{k\Delta x}{\epsilon}\right)\Delta x && \text{by Assumption 4.1.1} \\ &\leq \omega(0)\frac{\Delta x}{\epsilon} && \text{since } \omega \text{ is decreasing on } [0,1] \\ &= \omega(0)m^{-1} && \text{since } \epsilon = m\Delta x. \end{aligned} \tag{4.10}$$

So $w_k \leq \omega(0)m^{-1}$ implies $2w_k r_j^n \geq 2\omega(0)m^{-1}r_j^n$. In this case, we obtain

$$\begin{aligned} (w_{k-1} - w_k)\rho_{j+1}^n + 2w_k r_j^n &\geq cm^{-2}\rho_{\min} + 2\omega(0)m^{-1}r_j^n \\ &\geq cm^{-2}\rho_{\min} - 2\omega(0)m^{-1}L\Delta x \quad \text{by induction hypothesis} \\ &= cm^{-2}\rho_{\min} - 2\omega(0)m^{-1}L\epsilon m^{-1} \quad \text{since } \Delta x = \epsilon m^{-1} \\ &= m^{-2}(c\rho_{\min} - 2\omega(0)L\epsilon) \geq 0 \quad \text{since } c\rho_{\min} \geq 2\omega(0)L\epsilon \end{aligned}$$

Hence $p_{j,k}^n \geq 0$, and we conclude $c_{j,k}^n \geq 0$. □

4.1.3 Monotonicity of the update function $\mathcal{H}$

Having Lemma 4.1.6 at hand, we can finally deduce that the update function $\mathcal{H}$ given by (4.3) is nondecreasing in each of its arguments.

Proposition 4.1.7. *Under conditions of the Lemma 4.1.6, the update function $\mathcal{H}$ given by (4.3) is monotone.*

Proof. In Subsection 4.1.1, we have shown that $\mathcal{H}$ is nondecreasing with respect to ρ_{j-1}, ρ_j and ρ_{j+m}. Now consider $k = 1, 2, \ldots, m-1$. We have

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial \rho^n_{j+k}} &= \lambda\left(w_{k-1}\rho^n_j - w_k\rho^n_{j-1}\right)\\ &= \lambda\Big[(w_{k-1} - w_k)\rho^n_j + w_k(\rho^n_j - \rho^n_{j-1})\Big]\\ &= \lambda\Big[(w_{k-1} - w_k)\rho^n_j + w_k r^n_{j-1}\Big].\end{aligned}$$

Since $\lambda > 0$, it suffices to show that $(w_{k-1} - w_k)\rho^n_j + w_k r^n_{j-1} \geq 0$.

If $r^n_{j-1} \geq 0$, then we have

$$\begin{aligned}(w_{k-1} - w_k)\rho^n_j + w_k r^n_{j-1} &\geq cm^{-2}\rho_{\min} + w_k r^n_{j-1}\\ &\geq cm^{-2}\rho_{\min} \geq 0.\end{aligned}$$

If $r^n_{j-1} < 0$, then we have $w_k \leq \omega(0)m^{-1}$ by (4.10). This implies $w_k r^n_{j-1} \geq \omega(0)m^{-1}r^n_{j-1}$. Therefore, we obtain

$$\begin{aligned}(w_{k-1} - w_k)\rho^n_j + w_k r^n_{j-1} &\geq cm^{-2}\rho_{\min} + \omega(0)m^{-1}r^n_{j-1}\\ &\geq cm^{-2}\rho_{\min} - \omega(0)m^{-1}L\Delta x \qquad \text{by Lemma 4.1.6}\\ &= cm^{-2}\rho_{\min} - \omega(0)m^{-1}L\epsilon m^{-1} \qquad \text{since } \Delta x = \epsilon m^{-1}\\ &= m^{-2}\Big(c\rho_{\min} - \omega(0)L\epsilon\Big) \geq 0 \qquad \text{since } c\rho_{\min} \geq 2\omega(0)L\epsilon \geq \omega(0)L\epsilon\end{aligned}$$

Therefore, $\mathcal{H}$ is nondecreasing with respect to all its arguments, which means $\mathcal{H}$ is monotone. □

Summary

The maximum principle, Theorem 4.1.5, gives $\mathbf{L}^\infty$ bounds. The total variation estimates are obtained from the monotonicity of $\mathcal{H}$. The Oleinike's entropy condition can be obtained from a careful analysis on the one sided Lipschitz condition, see [8, LEMMA 4.3]. Thus, the numerical solutions computed by (4.3) is uniformly bounded in $\mathbf{BV}_{loc}(\mathbb{R} \times \mathbb{R}_+)$. So, Helly's theorem gives precompactness in $\mathbf{L}^1_{loc}$. Then, there exist a convergent subsequence in $\mathbf{L}^1_{loc}(\mathbb{R} \times \mathbb{R}_+)$ to a limit function ρ^* by Ascoli's theorem. Finally we can apply the Lax–Wendroff theorem to prove that the limit function ρ^* is the entropy solution of the local model, then by uniqueness of the entropy solution the whole sequence converge to the same limit function. That is, when we have

$$0 < \epsilon \leq \frac{c\rho_{\min}}{2L\omega(0)},$$

then the numerical solution converges to the unique entropy solution u of the local model (2.1) as both $\Delta x \to 0$ and $\epsilon \to 0$ simultaneously.

Chapter 5

Numerical Experiments

In Chapter 3, we developed the Lax–Friedrichs and Godunov schemes, including a complete algorithm for implementing the Lax–Friedrichs scheme (3.15). In Chapter 4, we provided the stability analysis of (3.14) and concluded that the method is asymptotically compatible. In the present chapter, we first present numerical experiments in Section 5.1 demonstrating that the choice of numerical quadrature weights w_k plays an essential role in the asymptotic compatibility of the numerical method (3.14). In Section 5.2, we then provide numerical evidence for the convergence of the method.

5.1 Comparison of Left endpoint and Normalized Left endpoint

We start by comparing the quadrature weights obtained from the left endpoint and the normalized left endpoint . For the Riemann problem (2.16), we choose $u_L = 0.6$ and $u_R = 0.1$ such that the problem becomes to solve

$$\begin{aligned} u_t + f(u)_x &= 0 \\ u(x,0) &= \begin{cases} 0.6 & \text{if } x < 0 \\ 0.1 & \text{if } x > 0. \end{cases} \end{aligned} \tag{5.1}$$

Since $u_L > u_R$, h in (2.18) equals

$$\inf\{g(u) \mid g \geq f \text{ and } g \text{ is concave on } [u_R, u_L]\},$$

since f is concave, h equals f itself. Thus, the entropy solution is given by

$$u(x,t) = \begin{cases} 0.6 & \text{for } x \leq -0.2t, \\ \dfrac{t-x}{2t} & \text{for } -0.2t \leq x \leq 0.8t, \\ 0.1 & \text{for } x \geq 0.8t. \end{cases} \tag{5.2}$$

Let us now choose $u_L = 0.1$ and $u_R = 0.6$ such that the problem becomes to solve

$$\begin{aligned} u_t + f(u)_x &= 0 \\ u(x,0) &= \begin{cases} 0.1 & \text{if } x < 0 \\ 0.6 & \text{if } x > 0. \end{cases} \end{aligned} \tag{5.3}$$

Since $u_L < u_R$, h in (2.18) equals

$$\sup\{g(u) \mid g \leq f \text{ and } g \text{ is convex on } [u_L, u_R]\},$$

since f is concave, h must equal a linear function. This leads us to a solution that consists of u_L and u_R, separated by a single shock with its speed s given by the Rankine–Hugoniot condition such that

$$\begin{aligned} s(t) &= \frac{f(u_R) - f(u_L)}{u_R - u_L} \\ &= \frac{0.6(1-0.6) - 0.1(1-0.1)}{0.6 - 0.1} \\ &= 0.3. \end{aligned}$$

Hence, the solution is given by

$$u(x,t) = \begin{cases} 0.1 & \text{if } x < 0.3t \\ 0.6 & \text{if } x > 0.3t. \end{cases} \tag{5.4}$$

We mention now that in all numerical experiments we do in this thesis, we choose the computational domain $[x_L, x_R] = [-2, 2]$, the linear kernel $\omega^\epsilon(y) = \frac{2(\epsilon - y)}{\epsilon^2}$ and the numerical viscosity $\alpha = 2$. So, we plot the rarefaction solution u given by (5.2) at final time $T = 1$ together with both the local and nonlocal numerical approximations.

Numerical Experiment 1a

We begin by implementing the Lax–Friedrichs type scheme (3.15) for $\epsilon = 0.1$ and $\Delta x = 0.02$

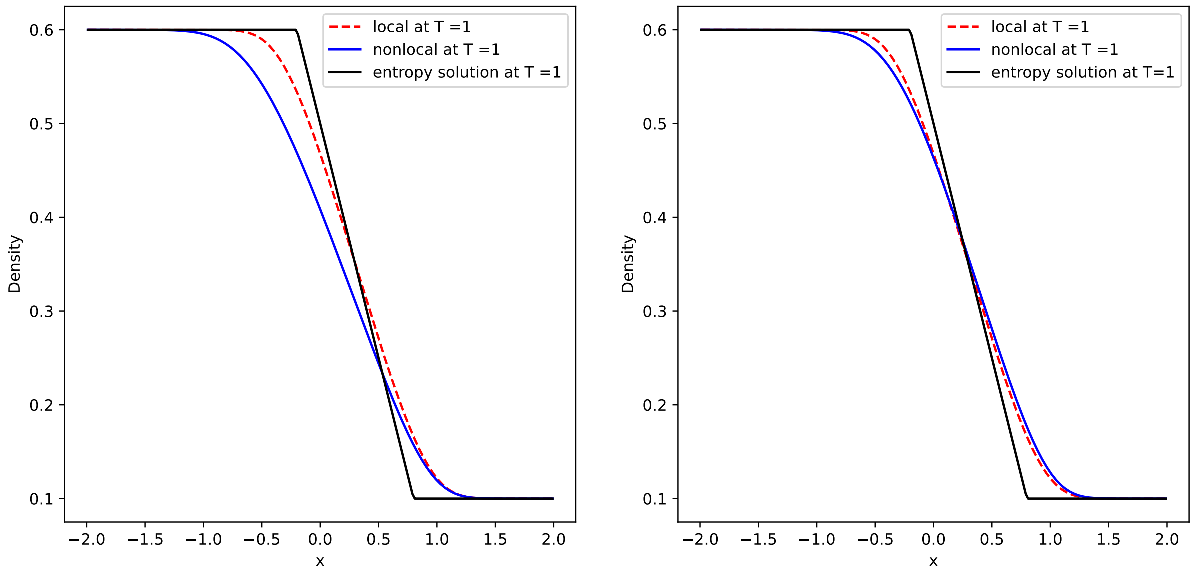

Figure 5.1: Numerical solutions and entropy solution at T = 1 corresponding to the left endpoint numerical quadrature weights (3.11) (left) and the normalized left endpoint numerical quadrature weights (3.12)(right).

As shown in Figure 5.1, the choice of numerical quadrature weights does not lead to a significant difference between the numerical solutions and the entropy solution of the local model (2.1). Nevertheless, a low level of accuracy remains at the shocks, where the numerical solutions deviate noticeably from the entropy solution. We aim to achieve convergence of the numerical solutions to the entropy solution as Δx and ϵ tend to zero simultaneously.

Numerical Experiment 1b

Next we choose $\epsilon = 0.01$ and $\Delta x = 0.005$

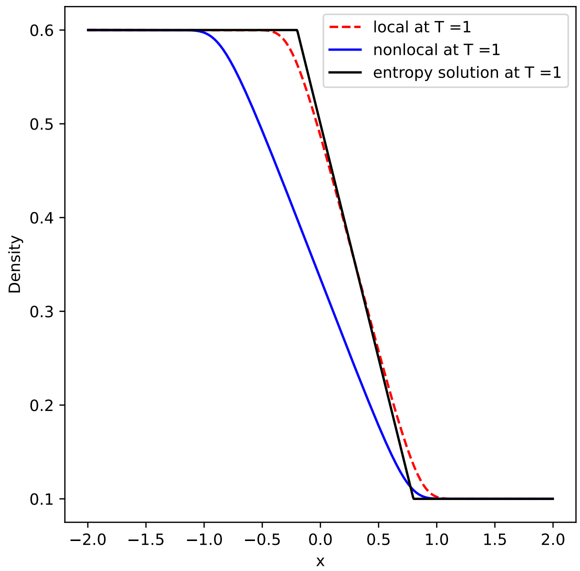

Figure 5.2: Numerical solutions and entropy solution at T = 1 corresponding to the left endpoint numerical quadrature weights (3.11) (left) and the normalized left endpoint numerical quadrature weights (3.12) (right).

As observed in Figure 5.2, contrary to our expectations, the left endpoint numerical quadrature weights cause the numerical solution of the nonlocal model to deviate further from both the numerical solution and the entropy solution of the local model. In contrast, the normalized left endpoint quadrature weights lead to improved accuracy and better alignment between the numerical solutions and the entropy solution.

Numerical Experiment 1c

Finally we take $\epsilon = 0.001$ and $\Delta x = 0.001$.

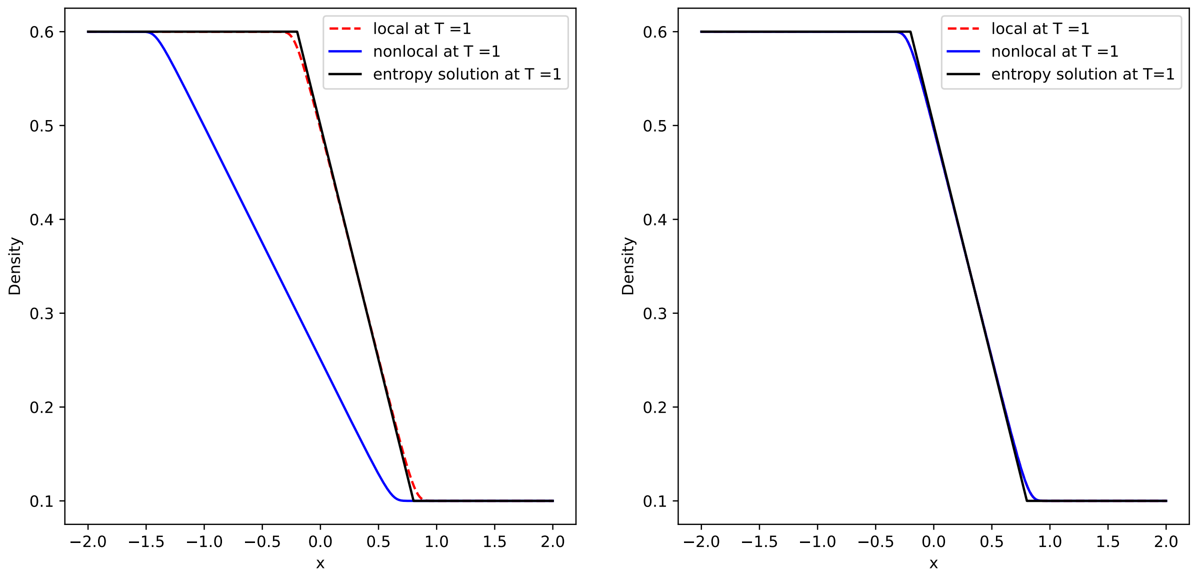

Figure 5.3: Numerical solutions and entropy solution at T = 1 corresponding to the left endpoint numerical quadrature weights (3.11) (left) and the normalized left endpoint numerical quadrature weights (3.12) (right).

We observe in Figure 5.3 that the numerical solution of the nonlocal model obtained using the left endpoint numerical quadrature weights completely fails. By contrast, significantly better accuracy is observed when using the normalized left endpoint numerical quadrature weights.

Summary

The numerical experiments above demonstrate that, as Δx and ϵ tend to zero simultaneously, while the Lax–Friedrichs type scheme based on the left endpoint quadrature weights becomes increasingly inaccurate, the normalized left endpoint quadrature weights consistently maintain stability and improved alignment with the entropy solution. The reason for this lies on the normalization condition, i.e, the sum of the numerical quadrature w_k equals 1. The above results provide evidence that the choice of numerical quadrature weights plays a crucial role in ensuring the asymptotic compatibility of the numerical method (3.14).

5.2 Numerical Convergence Analysis

The numerical experiments above indicate that the method (3.14) is consistent and stable when using the normalized left endpoint quadrature weights. We continue the numerical experiment to determine the convergence and order of convergence, and see if the Godunov numerical scheme(3.10) is more accurate than the Lax–Friedrichs scheme (3.15). We are interested in presenting numerical experiments on whether:

- $\rho^{\epsilon,\Delta x} \to u$ as both $\Delta x \to 0$ and $\epsilon \to 0$ simultaneously.
- $\rho^{\epsilon,\Delta x} \to \rho^{\epsilon}$ for fixed ϵ as $\Delta x \to 0$.

Considering the computed reference and numerical solutions as piecewise constant functions on each cell of size Δx, we measure the global error

$$e = U^{\text{ref}} - U^{\text{num}}$$

using the L^1 norm:

$$\|e\|_{L^1} = \Delta x \sum_{j=0}^{K-1} |e^j|.$$

Here, U^{ref} and U^{num} denote the reference solution and the numerical solution, respectively. We aim to obtain

$$\|e\|_{L^1} \to 0$$

as the parameters tend to zero.

5.2.1 Numerical Convergence to the Local Model

We will present the numerical experiments for $\rho^{\epsilon,\Delta x} \to u$ as both $\Delta x \to 0$ and $\epsilon \to 0$ simultaneously. In addition to the initial data (5.1) and (5.3), we also consider the smooth initial data

$$u(x, 0) = 0.5 + 0.3 \cos \frac{\pi x}{2}. \tag{5.5}$$

We remember that, as described in Chapter 2, we have a formula for solving Riemann problems of the local model. Hence for the Riemann initial data (5.1) and (5.3) we take the corresponding reference solution u to be the exact entropy solution given by (5.2) and (5.4), respectively. For the smooth initial data (5.5), we compute the corresponding reference solution u numerically on a fine grid for $\Delta x = \frac{4}{4096}$. With the relation $\epsilon = 3\Delta x$, we let both $\Delta x \to 0$ and $\epsilon \to 0$ simultaneously, although $\Delta x = \frac{1}{2^{4+m}}$ for $m = 0, 1, 2, 3, 4, 5, 6$, and compute the L^1 errors and order of convergence.

Focusing on the initial data given in (5.1), we plot the L^1 errors against the number of cells to compare the efficiency of the Godunov and Lax–Friedrichs scheme. We also present the observed order of convergence in Table 5.1

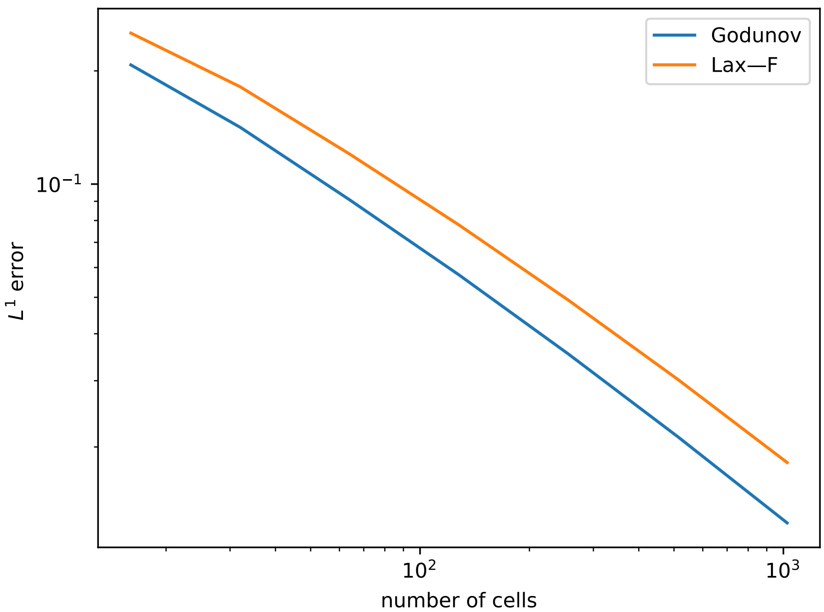

Figure 5.4: L^1 error vs. number of cells

Figure 5.4 indicates that the L^1-error for both Godunov and Lax–Friedrichs scheme tends to zero as the number of cells increases. In other words, both the schemes converge to the unique entropy solution u of the local model (2.1) as both $\Delta x \to 0$ and $\epsilon \to 0$ simultaneously. We observe as well that the Godunov scheme converges faster than Lax–Friedrichs scheme. A natural question is: how fast do the schemes converge? To answer this, we study the order of convergence r for both schemes.

We assume that the L^1 error on a mesh of size Δx_i can be written as

$$E_i = C(\Delta x_i)^r,$$

for some constant $C > 0$. If we have the error at two consecutive refinement levels (E_{i-1}, E_i), then we have

$$\frac{E_{i-1}}{E_i} = \left(\frac{\Delta x_{i-1}}{\Delta x_i}\right)^r.$$

Taking logarithms on both sides gives the standard expression for the order of convergence

$$r = \frac{\log(E_{i-1}) - \log(E_i)}{\log(\Delta x_{i-1}) - \log(\Delta x_i)}.$$

The order of convergence tells us how fast does a scheme converge to the correct solution, the higher order is the faster convergence.

Cells	L^1-errors	OOC
16	2.5205×10^{-1}	–
32	1.8141×10^{-1}	0.4744
64	1.2020×10^{-1}	0.5938
128	7.7851×10^{-2}	0.6267
256	4.9129×10^{-2}	0.6641
512	3.0257×10^{-2}	0.6993
1024	1.8184×10^{-2}	0.7346

(a) Lax–Friedrichs scheme

Cells	L^1-errors	OOC
16	2.0744×10^{-1}	–
32	1.4154×10^{-1}	0.5515
64	9.0816×10^{-2}	0.6402
128	5.7350×10^{-2}	0.6632
256	3.5334×10^{-2}	0.6987
512	2.1299×10^{-2}	0.7303
1024	1.2565×10^{-2}	0.7614

(b) Godunov scheme

Table 5.1: L^1-errors and observed order of convergence obtained using the Lax–Friedrichs scheme and the Godunov scheme in solving the local model with the Riemann initial data (5.1).

Table 5.1 presents the quantitative results for the L^1-errors and the observed order of convergence r for the Riemann initial data (5.1), where $u_L > u_R$. We observe that the L^1-errors produced by the Godunov scheme are consistently smaller than those obtained using the Lax–Friedrichs scheme. For both schemes, the observed order of convergence is at most 1, indicating relatively slow convergence.

Let us now consider the Riemann initial data (5.3), where $u_L < u_R$. We obtain the corresponding quantitative results:

Cells	L^1-errors	OOC
16	2.3182×10^{-1}	–
32	1.4869×10^{-1}	0.6407
64	8.8979×10^{-2}	0.7408
128	5.0150×10^{-2}	0.8272
256	2.6012×10^{-2}	0.9470
512	1.3128×10^{-2}	0.9866
1024	6.5769×10^{-3}	0.9971

(a) Lax–Friedrichs scheme

Cells	L^1-errors	OOC
16	1.2876×10^{-1}	–
32	8.5352×10^{-2}	0.5932
64	4.9126×10^{-2}	0.7969
128	2.6836×10^{-2}	0.8723
256	1.3336×10^{-2}	1.0089
512	6.7539×10^{-3}	0.9815
1024	3.3509×10^{-3}	1.0112

(b) Godunov scheme

Table 5.2: L^1-errors and observed order of convergence obtained using the Lax–Friedrichs scheme and the Godunov scheme in solving the local model with the Riemann initial data (5.3)

The observation in Table 5.2 indicates that the Godunov numerical scheme handles solutions with shocks effectively than The Lax–Friedrichs scheme. We also observe that for both schemes the order of convergence is 1.

Finally we repeat the above experiments with the smooth initial data (5.5), and obtain the corresponding quantitative results

Cells	L^1-errors	OOC
16	4.4055×10^{-1}	–
32	2.7531×10^{-1}	0.6783
64	1.6123×10^{-1}	0.7719
128	9.0595×10^{-2}	0.8316
256	4.9132×10^{-2}	0.8828
512	2.5445×10^{-2}	0.9493
1024	1.2221×10^{-2}	1.0580

(a) Lax–Friedrichs scheme

Cells	L^1-errors	OOC
16	3.1431×10^{-1}	–
32	1.8983×10^{-1}	0.7275
64	1.0959×10^{-1}	0.7926
128	6.1083×10^{-2}	0.8433
256	3.3016×10^{-2}	0.8876
512	1.7172×10^{-2}	0.9432
1024	8.4201×10^{-3}	1.0281

(b) Godunov scheme

Table 5.3: L^1-errors and observed order of convergence obtained using the Lax–Friedrichs scheme and the Godunov scheme in solving the local model with smooth initial data (5.5).

Summary

The experiments above provide evidence that the numerical schemes are asymptotically compatible, in the sense that they converge to the entropy solution of the local model (2.1) as both parameters Δx and ϵ tend to zero simultaneously. Moreover, although the stability analysis in Chapter 4 is established under the assumption that the initial data cannot contain negative jumps, the numerical experiments indicate that the methods remain robust even for the initial data (5.1), which contains a negative jump.

5.2.2 Numerical Convergence to the Nonlocal Model

For fixed ϵ, through numerical experiments we aim to deduce $\|e\|_{L^1} = \|\rho^\epsilon - \rho^{\epsilon,\Delta x}\|_{L^1} \to 0$ as $\Delta x \to 0$. We compute the reference solution ρ^ϵ numerically for $\Delta x = \frac{4}{4096}$ and $\epsilon = 0.1$. Holding $\epsilon = 0.1$ fixed and letting $\Delta x \to 0$, although $\Delta x = \frac{1}{2^{4+m}}$ for $m = 0, 1, 2, 3, 4, 5, 6$, we compute the L^1 errors and order of convergence.

Cells	L^1-errors	OOC
16	1.8626×10^{-1}	–
32	1.2159×10^{-1}	0.6154
64	7.4547×10^{-2}	0.7058
128	4.2751×10^{-2}	0.8022
256	2.3350×10^{-2}	0.8725
512	1.1829×10^{-2}	0.9811
1024	5.3589×10^{-3}	1.1423

(a) Lax–Friedrichs scheme

Cells	L^1-errors	OOC
16	1.2667×10^{-1}	–
32	7.3044×10^{-2}	0.7942
64	4.2832×10^{-2}	0.7701
128	2.3189×10^{-2}	0.8853
256	1.2184×10^{-2}	0.9285
512	5.9654×10^{-3}	1.0303
1024	2.6423×10^{-3}	1.1748

(b) Godunov scheme

Table 5.4: L^1-errors and observed order of convergence obtained using the Lax–Friedrichs scheme and the Godunov scheme in solving the nonlocal model with the Riemann initial data (5.1).

Cells	L^1-errors	OOC
16	2.0628×10^{-1}	–
32	1.0767×10^{-1}	0.9380
64	7.1225×10^{-2}	0.5962
128	3.8519×10^{-2}	0.8868
256	2.0491×10^{-2}	0.9106
512	1.0478×10^{-2}	0.9676
1024	4.8977×10^{-3}	1.0972

(a) Lax–Friedrichs scheme

Cells	L^1-errors	OOC
16	1.0844×10^{-1}	–
32	4.4663×10^{-2}	1.2798
64	3.3717×10^{-2}	0.4056
128	1.6571×10^{-2}	1.0248
256	8.2085×10^{-3}	1.0134
512	3.9258×10^{-3}	1.0641
1024	1.7225×10^{-3}	1.1885

(b) Godunov scheme

Table 5.5: L^1-errors and observed order of convergence obtained using the Lax–Friedrichs scheme and the Godunov scheme in solving the nonlocal model with the Riemann initial data (5.3)

Cells	L^1-errors	OOC
16	3.3092×10^{-1}	–
32	1.8065×10^{-1}	0.8733
64	9.9861×10^{-2}	0.8552
128	5.2189×10^{-2}	0.9362
256	2.6458×10^{-2}	0.9801
512	1.2647×10^{-2}	1.0649
1024	5.5133×10^{-3}	1.1978

(a) Lax–Friedrichs scheme

Cells	L^1-errors	OOC
16	1.8322×10^{-1}	–
32	8.6063×10^{-2}	1.0901
64	4.5197×10^{-2}	0.9291
128	2.2487×10^{-2}	1.0071
256	1.1061×10^{-2}	1.0237
512	5.1421×10^{-3}	1.1050
1024	2.2161×10^{-3}	1.2143

(b) Godunov scheme

Table 5.6: L^1-errors and observed order of convergence obtained using the Lax–Friedrichs scheme and the Godunov scheme in solving the nonlocal model with the smooth initial data (5.5).

Summary

The quantitative results above indicate that for fixed ϵ, $\rho^{\epsilon,\Delta x} \to \rho^{\epsilon}$ as $\Delta x \to 0$. In addition, the Godunov scheme is more efficient than the Lax–Friedrichs scheme. Based on our numerical experiments, the order of the developed numerical schemes is at most 1. Therefore, in the next chapter we will develop a second-order numerical scheme.

Part III

Second-order Numerical Methods

Chapter 6

A Second-Order Numerical Scheme

So far, we have considered first-order numerical methods. These methods are robust and can be applied to both the local and nonlocal models. However, their observed order of convergence is at most one. We therefore seek a second-order numerical method to improve computational efficiency.

6.1 Semidiscrete Formulation

The finite volume scheme (3.6) we stated in Chapter 3 is suitable for first order accurate schemes. In order to derive higher-order schemes, we need a formulation which is called a semidiscrete formulation. This formulation allows us to separately increase the order of spatial and temporal accuracy; see [10, Chapter 5].

Taking now each cell average over $\mathcal{C}_j$ to be given by

$$u_j(t) = \frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} u(x,t)\, dx \tag{6.1}$$

and integrating (2.1) over the domain $\mathcal{C}_j$

$$\int_{x_{j-1/2}}^{x_{j+1/2}} u_t + f(u)_x \, dx = 0$$

gives

$$\frac{d}{dt} u_j(t) + {}^{1}\!/\!{}_{\Delta x}\Big(f(u(x^-_{j+1/2}, t)) - f(u(x^+_{j-1/2}, t))\Big),$$

where

$$u(x^-_{j+1/2}, t) = \lim_{x \to x^-_{j+1/2}} u(x,t)$$

and

$$u(x^+_{j+1/2}, t) = \lim_{x \to x^+_{j+1/2}} u(x,t)$$

denotes the left and right-limit of u at $x_{j+1/2}$, respectively. Let F be an approximation of the fluxes such that,

$$F^{\pm}_{j+1/2} \approx f(u(x^{\pm}_{j+1/2}, t)),$$

then we obtain the semidiscrete formulation as

$$\begin{cases} \frac{d}{dt} u_j(t) + \frac{1}{\Delta x}\Big(F^-_{j+\frac{1}{2}} - F^+_{j-\frac{1}{2}}\Big) = 0 \\ u_j(0) = \frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} u_0(x)\, dx \end{cases} \tag{6.2}$$

Which is a system of ODEs that must be integrated in time by a suitable time integration routine.

6.1.1 Second Order Semidiscrete Scheme

Having the semidiscrete formulation (6.2), we can now separately increase the order of spatial and temporal accuracy. We observe in the formulation above that we need the edge values $u_j^\pm = u(x^\pm_{j+1/2}, t)$. For those values, we need to construct a linear piecewise function in each cell, so the edge values are evaluated on the reconstructed functions.

Given a cell average u_j^n of $\mathcal{C}_j$ at time t^n, we can construct a linear function

$$\bar{u}_j(x) = u_j^n + \sigma_j \frac{x - x_j}{\Delta x}, \quad x \in \mathcal{C}_j$$

where the slope σ_j is selected, such that $\bar{u}_j(x)$ is conservative and satisfy the TVD property. We choose the slope by the minmod limiter

$$\sigma_j^n = minmod\Big(u_j^n - u_{j-1}^n,\, u_{j+1}^n - u_j^n\Big)\,,$$

here the function minmod is defined by

$$minmod(a_1, \ldots, a_m) = \begin{cases} \operatorname{sgn}(a_1) \min_{1 \le \ell \le m} |a_\ell|, & \text{if } \operatorname{sgn}(a_1) = \cdots = \operatorname{sgn}(a_m), \\ 0, & \text{otherwise.} \end{cases}$$

We can now use $\bar{u}_j$ to approximate $u(x^-_{j+1/2}, t)$ and $u(x^+_{j+1/2}, t)$, such that the left and right-limit values are given by

$$u^-_{j+\frac{1}{2}} := \lim_{x \to x^-_{j+1/2}} \bar{u}_j(x) = u_j^n + \frac{1}{2}\sigma_j^n$$

and

$$u^+_{j+\frac{1}{2}} := \lim_{x \to x^+_{j+1/2}} \bar{u}_j(x) = u_{j+1}^n - \frac{1}{2}\sigma_{j+1}^n,$$

respectively.

In Chapter 3, we approximated the integral term

$$\int_0^\epsilon \omega^\epsilon(y)\rho(x+y,t)\,dy \tag{6.3}$$

using piecewise constants, which led us to the first-order finite volume scheme. In order to obtain the second-order finite volume scheme, let $\bar{\rho}_j(x,t)$ be a linear reconstructed function corresponding to the nonlocal model. Then, we approximate the integral term employing the piecewise linear function $\bar{\rho}(x,t) = \bar{\rho}_j(x)$ for $x \in \mathcal{C}_j$ and the trapezoidal rule. Following the approach in [6], let

$$M := \lceil \frac{\epsilon}{\Delta x} \rceil, \qquad \text{denotes the number of cells involved in (6.3),}$$

and define

$$I_{j+1/2} := \int_0^\epsilon \omega^\epsilon(y)\rho(x_{j+1/2}+y,t)\,dy.$$

We apply first $\bar{\rho}$ such that $\rho(x_{j+1/2}+y,t) \approx \bar{\rho}(x_{j+1/2}+y,t)$ to get

$$\begin{aligned} I_{j+1/2} &\approx \int_0^\epsilon \omega^\epsilon(y)\bar{\rho}(x_{j+1/2}+y,t)\,dy \\ &= \sum_{k=0}^{M-1} \int_{k\Delta x}^{(k+1)\Delta x} \omega^\epsilon(y)\bar{\rho}(x_{j+1/2}+y,t)\,dy. \end{aligned}$$

We then apply the trapezoidal rule to get

$$
\begin{aligned}
I_{j+1/2} &\approx \frac{\Delta x}{2}\sum_{k=0}^{M-1}\Big(\omega^\epsilon(k\Delta x)\bar\rho(x_{j+1/2}+k\Delta x,t)+\omega^\epsilon((k+1)\Delta x)\bar\rho(x_{j+1/2}+(k+1)\Delta x,t)\Big)\\
&= \frac{\Delta x}{2}\sum_{k=0}^{M-1}\Big(\omega^\epsilon(k\Delta x)\bar\rho(x_{j+k+1/2},t)+\omega^\epsilon((k+1)\Delta x)\bar\rho(x_{j+k+1+1/2},t)\Big)\\
&= \frac{\Delta x}{2}\sum_{k=0}^{M-1}\Big(W_k\bar\rho(x_{j+k+1/2},t)+W_{k+1}\bar\rho(x_{j+k+3/2},t)\Big)\\
&= \frac{\Delta x}{2}\sum_{k=0}^{M-1}\Big(W_k\rho^+_{j+k+1/2}+W_{k+1}\rho^-_{j+k+3/2}\Big)
\end{aligned}
$$

where W_k is given by

$$W_k = \omega^\epsilon(k\Delta x), \qquad k = 0,\cdots M.$$

We define the numerical quadrature $\tilde W_k$

$$Q_{\Delta x} := \frac{\Delta x}{2}\sum_{k=0}^{M-1}\Big(W_k+W_{k+1}\Big), \quad \text{and } \tilde W_k := \frac{W_k}{Q_{\Delta x}}, \qquad k = 0,\cdots M,$$

such that the normalization condition is satisfied, in the sense

$$\frac{\Delta x}{2}\sum_{k=0}^{M-1}\Big(\tilde W_k+\tilde W_{k+1}\Big) = 1.$$

If we now replace W_k by $\tilde W_k$ in

$$R_{j+1/2} := \frac{\Delta x}{2}\sum_{k=0}^{M-1}\Big(W_k\rho^+_{j+k+1/2}+W_{k+1}\rho^-_{j+k+3/2}\Big),$$

we obtain

$$R_{j+1/2} = \frac{\Delta x}{2}\sum_{k=0}^{M-1}\Big(\tilde W_k\rho^+_{j+k+1/2}+\tilde W_{k+1}\rho^-_{j+k+3/2}\Big).$$

Now let the numerical flux $F^-_{j+1/2}$ in (6.2) be defined by the Godunov-type flux

$$F^-_{j+1/2} := \rho^-_{j+1/2}(1-R_{j+1/2}),$$

and denote

$$L_\epsilon(\rho)_j = \frac{1}{\Delta x}(F^-_{j+1/2}-F^+_{j-1/2}).$$

Then, we obtain the nonlocal semidiscrete scheme:

$$
\begin{cases}
\dfrac{d}{dt}\rho_j(t)+L_\epsilon(\rho)_j = 0\\
\rho_j(0) = \dfrac{1}{\Delta x}\displaystyle\int_{x_{j-1/2}}^{x_{j+1/2}}\rho_0(x)\,dx
\end{cases}
\tag{6.4}
$$

Proposition 6.1.1. *Suppose ω^ϵ has compact support in $[0,\epsilon]$. Let $\Delta x > 0$ be fixed. Then, as $\epsilon \to 0$, the nonlocal semidiscrete scheme (6.4) converges to the local semidiscrete scheme given by*

$$
\begin{cases}
\dfrac{d}{dt}u_j(t)+L(u)_j = 0,\\
u_j(0) = \dfrac{1}{\Delta x}\displaystyle\int_{x_{j-1/2}}^{x_{j+1/2}}u_0(x)\,dx,
\end{cases}
\tag{6.5}
$$

where

$$L(u)_j := \frac{1}{\Delta x}(G^-_{j+1/2} - G^+_{j-1/2}), \qquad G^-_{j+1/2} := u^-_{j+1/2}(1 - u^+_{j+1/2})$$

Proof. It suffices to prove that

$$R_{j+\frac{1}{2}} \to \rho^+_{j+\frac{1}{2}} \qquad \text{as } \epsilon \to 0,$$

since then we get

$$F^-_{j+\frac{1}{2}} = \rho^-_{j+1/2}(1 - \rho^+_{j+1/2}),$$

which has the same form as

$$G^-_{j+1/2} = u^-_{j+1/2}(1 - u^+_{j+1/2}).$$

We keep the convention that u denotes the local solution and ρ denotes the nonlocal solution. If Δx is fixed, and $\epsilon \to 0$, sooner or later we get $\epsilon < \Delta x$. In that case $M = 1$. Hence we get

$$Q_{\Delta x} = \frac{\Delta x}{2}\Big(W_0 + W_1\Big), \qquad \tilde{W}_0 = \frac{W_0}{Q_{\Delta x}}, \qquad \tilde{W}_1 = \frac{W_1}{Q_{\Delta x}},$$

and

$$R_{j+1/2} = \frac{\Delta x}{2}\Big(\tilde{W}_0 \rho^+_{j+1/2} + \tilde{W}_1 \rho^-_{j+3/2}\Big).$$

Since ω^ϵ has compact support in $[0, \epsilon]$, we have

$$W_1 = \omega^\epsilon(\Delta x) = 0 \text{ and } \tilde{W}_1 = 0.$$

So

$$R_{j+1/2} = \frac{\Delta x}{2}\tilde{W}_0 \rho^+_{j+1/2},$$

and

$$Q_{\Delta x} = \frac{\Delta x}{2} W_0, \qquad \tilde{W}_0 = \frac{W_0}{Q_{\Delta x}} = \frac{2}{\Delta x}.$$

Hence

$$R_{j+\frac{1}{2}} = \rho^+_{j+\frac{1}{2}}.$$

Therefore, $F^-_{j+1/2} \to G^-_{j+1/2}$ and

$$L_\epsilon(\rho)_j \to L(u)_j \qquad \text{as } \epsilon \to 0.$$

This proves the claim. □

6.1.2 Second Order Godunov Scheme

Let us apply now the forward Euler method to the nonlocal semidiscrete (6.4) to obtain

$$\rho^{n+1}_j = \rho^n_j - \Delta t L_\epsilon(\rho^n)_j.$$

It turns out this scheme is only first-order accurate, even though we have used the piecewise linear reconstruction in space; see [10, p. 76]. In order to obtain a second-order accurate scheme, we have to use the so-called strong stability preserving (SSP) Runge-Kutta methods. Hence we need to perform the following steps

$$\begin{aligned} \rho^{(1)}_j &= \rho^n_j - \Delta t L_\epsilon(\rho^n)_j \\ \rho^{(2)}_j &= \rho^{(1)}_j - \Delta t L_\epsilon(\rho^{(1)})_j \\ \rho^{n+1}_j &= \frac{1}{2}(\rho^n_j + \rho^{(2)}_j) \end{aligned} \tag{6.6}$$

to obtain the second-order Godunov scheme.

Chapter 7

Comparison of First and Second-Order Godunov Scheme

In this chapter, we will present numerical experiments where we compare the first- and second-order Godunov schemes.

7.1 Numerical Convergence to the local Model

Cells	**L^1-errors**	**OOC**
16	2.0744×10^{-1}	–
32	1.4154×10^{-1}	0.5515
64	9.0816×10^{-2}	0.6402
128	5.7350×10^{-2}	0.6632
256	3.5334×10^{-2}	0.6987
512	2.1299×10^{-2}	0.7303
1024	1.2565×10^{-2}	0.7614

(a) First-order Godunov scheme

Cells	**L^1-errors**	**OOC**
16	1.5983×10^{-1}	–
32	1.0004×10^{-1}	0.6759
64	6.0907×10^{-2}	0.7159
128	3.6599×10^{-2}	0.7348
256	2.1605×10^{-2}	0.7604
512	1.2515×10^{-2}	0.7877
1024	7.0899×10^{-3}	0.8199

(b) Second-order Godunov scheme

Table 7.1: L^1-errors and observed order of convergence obtained using the first- and second-order Godunov schemes in solving the local model with the Riemann initial data (5.1).

Cells	**L^1-errors**	**OOC**
16	1.2876×10^{-1}	–
32	8.5352×10^{-2}	0.5932
64	4.9126×10^{-2}	0.7969
128	2.6836×10^{-2}	0.8723
256	1.3336×10^{-2}	1.0089
512	6.7539×10^{-3}	0.9815
1024	3.3509×10^{-3}	1.0112

(a) First-order Godunov scheme

Cells	**L^1-errors**	**OOC**
16	9.8507×10^{-2}	–
32	6.1071×10^{-2}	0.6897
64	2.9665×10^{-2}	1.0417
128	1.5495×10^{-2}	0.9370
256	7.7365×10^{-3}	1.0021
512	3.3069×10^{-3}	1.2262
1024	1.6722×10^{-3}	0.9838

(b) Second-order Godunov scheme

Table 7.2: L^1-errors and observed order of convergence obtained using the first- and second-order Godunov schemes in solving the local model with the Riemann initial data (5.3).

Cells	L^1-errors	OOC
16	3.1431×10^{-1}	–
32	1.8983×10^{-1}	0.7275
64	1.0959×10^{-1}	0.7926
128	6.1083×10^{-2}	0.8433
256	3.3016×10^{-2}	0.8876
512	1.7172×10^{-2}	0.9432
1024	8.4201×10^{-3}	1.0281

(a) First-order Godunov scheme

Cells	L^1-errors	OOC
16	2.3215×10^{-1}	–
32	1.3042×10^{-1}	0.8319
64	7.2110×10^{-2}	0.8549
128	3.9305×10^{-2}	0.8755
256	2.1099×10^{-2}	0.8975
512	1.1225×10^{-2}	0.9104
1024	5.8947×10^{-3}	0.9292

(b) Second-order Godunov scheme

Table 7.3: L^1-errors and observed order of convergence obtained using the first- and second-order Godunov schemes in solving the local model with the smooth initial data (5.5).

The numerical experiments above indicate that the second-order scheme (6.6) is asymptotically compatible. However, it does not seem that the L^1-errors and observed order of convergence have improved.

7.2 Numerical Convergence to the Nonlocal Model

We do now the comparison on the nonlocal model (2.2).

Cells	L^1-errors	OOC
16	1.2667×10^{-1}	–
32	7.3044×10^{-2}	0.7942
64	4.2832×10^{-2}	0.7701
128	2.3189×10^{-2}	0.8853
256	1.2184×10^{-2}	0.9285
512	5.9654×10^{-3}	1.0303
1024	2.6423×10^{-3}	1.1748

(a) First-order Godunov scheme

Cells	L^1-errors	OOC
16	5.0110×10^{-2}	–
32	1.7309×10^{-2}	1.5336
64	7.9544×10^{-3}	1.1217
128	3.0467×10^{-3}	1.3845
256	1.4243×10^{-3}	1.0971
512	6.5964×10^{-4}	1.1104
1024	2.9541×10^{-4}	1.1590

(b) Second-order Godunov scheme

Table 7.4: L^1-errors and observed order of convergence obtained using the first- and second-order Godunov schemes in solving the nonlocal model with the Riemann initial data (5.1).

Cells	L^1-errors	OOC
16	1.0844×10^{-1}	–
32	4.4663×10^{-2}	1.2798
64	3.3717×10^{-2}	0.4056
128	1.6571×10^{-2}	1.0248
256	8.2085×10^{-3}	1.0134
512	3.9258×10^{-3}	1.0641
1024	1.7225×10^{-3}	1.1885

(a) First-order Godunov scheme

Cells	L^1-errors	OOC
16	7.1040×10^{-2}	–
32	1.8091×10^{-2}	1.9734
64	1.5403×10^{-2}	0.2321
128	5.7654×10^{-3}	1.4177
256	1.9766×10^{-3}	1.5444
512	5.6019×10^{-4}	1.8190
1024	1.4551×10^{-4}	1.9448

(b) Second-order Godunov scheme

Table 7.5: L^1-errors and observed order of convergence obtained using the first- and second-order Godunov schemes in solving the nonlocal model with the Riemann initial data (5.3).

Cells	L^1-errors	OOC
16	1.8322×10^{-1}	–
32	8.6063×10^{-2}	1.0901
64	4.5197×10^{-2}	0.9291
128	2.2487×10^{-2}	1.0071
256	1.1061×10^{-2}	1.0237
512	5.1421×10^{-3}	1.1050
1024	2.2161×10^{-3}	1.2143

(a) First-order Godunov scheme

Cells	L^1-errors	OOC
16	7.5224×10^{-2}	–
32	2.2102×10^{-2}	1.7670
64	6.3384×10^{-3}	1.8020
128	2.1914×10^{-3}	1.5322
256	4.8969×10^{-4}	2.1619
512	1.5326×10^{-4}	1.6759
1024	3.0656×10^{-5}	2.3218

(b) Second-order Godunov scheme

Table 7.6: L^1-errors and observed order of convergence obtained using the first- and second-order Godunov schemes in solving the nonlocal model with the smooth initial data (5.5).

We observe in the above results that both the L^1-errors and observed order of convergence are improved.

7.3 Conclusion

The second-order numerical method for the nonlocal model (2.2) was developed in [6]. However, the authors do not address asymptotic compatibility. Moreover, their BV estimate in space blows up as $\Delta x \to 0$ and $\epsilon \to 0$. This motivates our investigation of the asymptotic compatibility problem.

Our first positive result was the convergence of the nonlocal-to-local limit, summarized in Proposition 6.1.1. To prove asymptotic compatibility, it may be useful to investigate whether the second-order scheme satisfies a one-sided Lipschitz condition analogous to that established for first-order schemes in Lemma 4.1.6. Though, the proof of Lemma 4.1.6 is not rigorous.

Bibliography

[1] Sebastien Blandin and Paola Goatin. 'Well-Posedness of a Conservation Law with Non-Local Flux Arising in Traffic Flow Modeling'. In: *Numerische Mathematik* 132.2 (1st Feb. 2016), pp. 217–241. ISSN: 0945-3245. DOI: 10.1007/s00211-015-0717-6. URL: https://doi.org/10.1007/s00211-015-0717-6 (visited on 26/04/2026).

[2] Alberto Bressan and Wen Shen. 'Entropy Admissibility of the Limit Solution for a Nonlocal Model of Traffic Flow'. In: *Communications in Mathematical Sciences* 19.5 (2021), pp. 1447–1450. ISSN: 15396746, 19450796. DOI: 10.4310/CMS.2021.v19.n5.a12. URL: https://link.intlpress.com/JDetail/1806261857369411585 (visited on 14/05/2026).

[3] Giuseppe Maria Coclite et al. 'A General Result on the Approximation of Local Conservation Laws by Nonlocal Conservation Laws: The Singular Limit Problem for Exponential Kernels'. In: *Annales de l'Institut Henri Poincaré C, Analyse non linéaire* 40.5 (4th Nov. 2022), pp. 1205–1223. ISSN: 0294-1449, 1873-1430. DOI: 10.4171/aihpc/58. URL: https://ems.press/doi/10.4171/aihpc/58 (visited on 14/05/2026).

[4] Maria Colombo et al. 'Local Limit of Nonlocal Traffic Models: Convergence Results and Total Variation Blow-Up'. In: *Annales de l'Institut Henri Poincaré C, Analyse non linéaire* 38.5 (Oct. 2021), pp. 1653–1666. ISSN: 0294-1449, 1873-1430. DOI: 10.1016/j.anihpc.2020.12.002. arXiv: 1808.03529 [math]. URL: http://arxiv.org/abs/1808.03529 (visited on 11/05/2026).

[5] Paola Goatin. 'Conservation Laws with Local Flux Constraints Arising in Traffic Flow Modeling'. In: *ESAIM: Proceedings and Surveys* 45 (Sept. 2014). Ed. by Jean-Stéphane Dhersin, pp. 8–17. ISSN: 2267-3059. DOI: 10.1051/proc/201445002. URL: http://www.esaim-proc.org/10.1051/proc/201445002 (visited on 13/05/2026).

[6] Veerappa Gowda G. D., Sudarshan Kumar Kenettinkara and Nikhil Manoj. 'Convergence of a Second-Order Scheme for Non-Local Conservation Laws'. In: *ESAIM: Mathematical Modelling and Numerical Analysis* 57.6 (Nov. 2023), pp. 3439–3481. ISSN: 2822-7840, 2804-7214. DOI: 10.1051/m2an/2023080. URL: https://www.esaim-m2an.org/10.1051/m2an/2023080 (visited on 10/05/2026).

[7] Helge Holden and Nils Henrik Risebro. *Front Tracking for Hyperbolic Conservation Laws*. Vol. 152. Applied Mathematical Sciences. Berlin, Heidelberg: Springer Berlin Heidelberg, 2015. ISBN: 978-3-662-47506-5 978-3-662-47507-2. DOI: 10.1007/978-3-662-47507-2. URL: http://link.springer.com/10.1007/978-3-662-47507-2 (visited on 10/05/2026).

[8] Kuang Huang and Qiang Du. 'Asymptotic Compatibility of a Class of Numerical Schemes for a Nonlocal Traffic Flow Model'. In: *SIAM Journal on Numerical Analysis* 62.3 (30th June 2024), pp. 1119–1144. ISSN: 0036-1429, 1095-7170. DOI: 10.1137/23M154488X. URL: https://epubs.siam.org/doi/10.1137/23M154488X (visited on 26/04/2026).

[9] Alexander Keimer and Lukas Pflug. 'Existence, Uniqueness and Regularity Results on Nonlocal Balance Laws'. In: *Journal of Differential Equations* 263.7 (5th Oct. 2017), pp. 4023–4069. ISSN: 0022-0396. DOI: 10.1016/j.jde.2017.05.015. URL: https://www.sciencedirect.com/science/article/pii/S0022039617302759 (visited on 10/05/2026).

[10] Siddhartha Mishra, Ulrik Skre Fjordholm and Remi Abgrall. *Numerical Methods for Conservation Laws and Related Equations*. University of Oslo, Department of Mathematics. URL: https://www.mn.uio.no/math/english/people/aca/ulriksf/lecture-notes/Numerical%20methods%20for%20hyperbolic%20equations.pdf (visited on 26/04/2026).

[11] Nicola De Nitti and Kuang Huang. *Asymptotically Compatible Entropy-Consistent Discretization for a Class of Nonlocal Conservation Laws*. 30th Sept. 2025. DOI: 10.48550/arXiv.2510.00221. arXiv: 2510.00221 [math]. URL: http://arxiv.org/abs/2510.00221 (visited on 11/05/2026). Pre-published.

[12] Xiaochuan Tian and Qiang Du. 'Asymptotically Compatible Schemes and Applications to Robust Discretization of Nonlocal Models'. In: *SIAM Journal on Numerical Analysis* 52.4 (Jan. 2014), pp. 1641–1665. ISSN: 0036-1429, 1095-7170. DOI: 10.1137/130942644. URL: http://epubs.siam.org/doi/10.1137/130942644 (visited on 09/05/2026).